Who says science isn't exciting?

An episode by episode breakdown of
the alien science in Roswell, New Mexico

by thesquidkid (Mil)

Table of contents

INTRODUCTION

Have you ever wondered, as you were watching this amazing little alien show, what the science was all about? What all the complicated scientific terms meant? Whether or not the writers researched what was going on? Well I have, and this essay might be for you! And don't worry, even if you know nothing about science, I will try to make it as clear as possible.

This is more or less a corrected, more readable version of my science meta, where I debunk the science episode by episode. In this paper, everything will be a put a bit more together, with still I hope a clear structure. The idea behind this PDF is that I will try to write it as close as possible to an actual scientific article, in what I hope is a good exercise for my thesis. I also hope that it is better organised, and more readable than what I had originally written on Tumblr. I will also keep updating this pdf with every new science meta I make, and everything should be organised nicely, with cross-references and an ability to search through for certain words/notions from the show.

This book is structured in what I hope is a rather straightforward way. Each chapter will represent a season, and each section a different episode. Furthermore, there will be some definitions scattered around, that I will reference back to when needed. If you click on any words in purple, it will lead you to the definition of that word, that was given somewhere earlier, or to a link to what is specifically mentioned, e.g. in the case of the Tumblr post referenced above (no rickrolls in this paper, I swear). Numbers in purple refer to either definitions numbers or episodes, where you can also click on it to be directed there. If the number in purple is in between square brackets, i.e. [1], then it will lead you to the page of references, where you can access the article/website I used to gather the information. Please note that this is not an academic paper, as I am citing Wikipedia. I have written academic papers (and I will keep doing that for a while), this is simply a way for me to have fun doing research, and to keep practicing my LaTeX skills.

An index is given at the very end, for an easier navigation, and a list of all dialogues that I have analysed is right after the introduction. These should hopefully make it easier to find the exact scientific stuff you need! Note that for the index, there are some words that were made up by the show writers (for instance, the source of many of my headaches: butyricol). They are listed in the index with an asterisk, e.g. as butyricol*.

Disclaimers

I am not American and English is not my first language. Hence things that may seem obvious to other people, may not be that obvious to me, and vice versa. I apologise in advance for any mistakes made in the document, I tried to correct as much as I could, but some may have slipped through. Please keep in mind that

I was mostly sleep deprived when I wrote the original meta, so if things aren't understandable, feel free to reach out.

Secondly, I started this project in the second year of my bachelor's in applied mathematics, and ended it right at the end of my master's in the same field. I stopped biology and chemistry in the middle of high school, and even before then, my grades were not that great. However, I am a huge science nerd, and occasionally watch hours long YouTube videos and read obscure articles on field that could not be further than mine. There may be some inaccuracies in what I wrote, namely for biology and chemistry facts, I tried to compare articles and sources to give you the best definition, but some terms were out of my grasp.

I would also like to add that despite loving all types of science, and enjoying working on this project, I have a clear preference for maths, so there may be a few fangirl moments here and there. I will however not apologise for being weird.

Lastly, this show is fictional. I mean, it's about aliens who look a lot like humans, and who can move stuff with their brains, heal, cause blackouts by being angry or turned on, or get inside people's minds. So sometimes the science isn't really accurate. That's okay, the writers are show writers, not doctors who wrote a thesis a particular subject. And besides, if it was about actual science, it won't be as fun!

SEASON 1

1.1 PILOT

The pilot episode, and with it the show, opens with Liz talking,

> *"Every small town has a story, but my hometown has a legend. Roswell was a sleepy cowboy settlement in postwar America, full of farmers and military men, until, one day, something extraordinary happened. Or so the legend goes. Ever since, UFO enthusiasts have flooded in, searching for some cosmic phenomenon to prove we're not all alone in the universe."*

The first element of science is one which lays at the foundation of the show: we are not alone in the universe. This question is as old as mankind, and has faced quite a few debates, so I won't really dive deeply into it, especially since in this show aliens are clearly considered fiction. However, from a physical, astronomical and biological point of view, we are indeed not alone in the universe. As of today, we know the existence of at least 3850 planets around other stars and it is statistically near impossible that Earth is the only one that can harbour life. And just like Liz, I believe in statistics.

Then, at the high school reunion, Alex tells Michael

> *"I'm serious. My chemical engineers found high levels of phenyl-2-propanone around your Airstream."*

Chemistry was not my strong suit in high school, so I had to do some rather extensive research, and I could still be very wrong. Below is the definition of phenyl-2-propanone, and the components that compose it:

DEFINITION 1.1.1: PHENYL-2-PROPANONE

Phenyl-2-propanone, often times abbreviated as P2P, and more commonly known as "phenylacetone", is a colorless oil that is soluble in organic solvents. In the United States, it is considered a level II controlled substance (basically it's not illegal, but it's use is controlled). It is created in the human body during effort (albeit in very low quantity), namely in the creation of *amphetamine* and *methamphetamine* (which can also be fabricated artificially) [1, 2].

DEFINITION 1.1.2: AMPHETAMINE

Amphetamine is a central nervous system stimulant that causes euphoria and improves cognitive control, while also increasing muscle strength. As a drug, it is prescribed for the treatment of ADHD, but is also a recreational drug that can lead to addiction and psychosis [3].

DEFINITION 1.1.3: METHAMPHETAMINE

Methamphetamine is a similar drug (where the name meth comes from) to amphetamine, with a higher risk of addiction [4].

Michael responds by denying that it was P2P. The next part is purely theoretical on my side, as we are not given much more[1]. This is where my hypothesis comes into play: acetone. We know that the aliens drink acetone, that it's like a painkiller for them. But what if it was more than that? What if pieces of alien tech had the same component as acetone (namely 2-Phenyl-2-Propanone)? We know that Michael had pieces of tech in his trailer, that he had managed to extract some of the fluids from the pods. It would only make sense that traces of 2-Phenyl-2-Propanone would still be around his airstream, and why Alex's engineers assumed it was meth (possibly from the concentration of it, which is higher in meth than in nail dissolvent for instance)[2].

1.2 SO MUCH FOR THE AFTERGLOW

Liz mentions her research to Max:

> *"I'm a scientist, Max. It's what I do. I got a grant for my vascular regeneration study."*

In short, Liz's research is about restoring normal vascular functions, and the growth of new blood vessels[3] [5]. Vascular refers here to the veins, arteries, and all through which blood flows in a body.

1.3 TEARIN' UP MY HEART

> *"Chronic low-level anxiety keeps the hypothalamic pituitary adrenal axis activated"*

This is what Liz says when she starts her experiment on Max. Lots of complicated scientific words. Let's break them down, shall we?

DEFINITION 1.3.1: CHRONIC LOW-LEVEL ANXIETY

Chronic low-level anxiety: chronic means persisting for a long time or constantly recurring. Anxiety is a feeling of unease such as worry or fear.

[1] From the script drafts, we can see that Michael also mentions something about the 21st component being different, but I will get back to that later on.

[2] N.B. At the time I wrote this, I wasn't aware of the notion of the 21st component in the first scripts, and therefore could only speculate. However, this simple sentence about that unknown component makes my theory even stronger, as you will see later on.

[3] I will go further into Liz's research later on, as the show dives deeper into it

DEFINITION 1.3.2: HYPOTHALAMIC-PITUITARY-ADRENAL AXIS

The **hypothalamic-pituitary-adrenal axis** (HPA) is a set of direct influences and feedback interactions between the *hypothalamus*, the *pituitary gland* and the *adrenal glands* [6].
The **hypothalamus** is a part of the brain that controls body temperature, hunger, thirst, fatigue, and sleep [7].
The **pituitary gland** is a gland the size of a pea that produces hormones which in turn regulate the growth of the human body, blood pressure, energy management, the sex organs, the thyroid glands, and some aspects of pregnancy, childbirth and breastfeeding, as well as the water/salt concentration in the kidneys and pain relief [8].
The **adrenal glands** produce adrenaline, and regulate the metabolism and the immune system [9].

Basically, what Liz is saying, is that chronic anxiety (when it is not at a worryingly high level) keeps the (human) body regulated, as it supposed to be. So, the fact that Max seems to be extra stressed would shift his hormone levels, leading to his body not being normally regulated – i.e. raised body temperature, fits of anger, higher blood pressure, extra adrenaline, etc. All of which are symptoms Max seems to be having following Liz's resurrection.

Then, Max explains the effects that acetone has on them. Since this is all hand-wave science (not criticising, hand-wave science is really cool in science fiction), I won't really detail on that. Just remember kids, drinking acetone isn't good for your health!

The next bit of science is scattered a little through the episode, so I've gathered all the mentions of electricity and such below. All lines are spoken by Liz.

> *"Prediction: Max's ability to manipulate electromagnetic energy gives him the power to heal and also to harm."*
>
> *"I've established your baseline. Now I'm monitoring your electromagnetic output and how it correlates with... emotional triggers."*
>
> *"Okay, are you all capable of electrostatic discharge... Releasing energy intentionally?"*
>
> *"In nature, there are animals... Plants, even... That do variations of this."*
>
> *"So if you can heal with electricity, can you also harm?"*

Liz mentions animals and plants releasing electricity as a defense mechanism. This is a real fact (so cool, right?!), for instance, the electric eel can generate their own electricity. We call animals who generate electricity **electrogenic animals**, and those who can detect electricity **electroreceptive**.

Most animals are electroreceptive, like the platypus, bees, sharks, dolphins, some types of bacteria, yeast, etc. And as for electrogenic animals, there are about 350 species of fish that can generate electric rays, which can be useful in the deep ocean as it is usually very dark. They also use it to attack their preys. The most fascinating part here is that electrogenic fishes range from a power of 1V to 860V (for reference, an AA battery generates 1.5V). The electric eel can produce rays of 860V, which as you can imagine, is pretty deadly for a human, but of course, the eel does not electrocute itself, as that might be quite inconvenient[4]. And as for plants, scientists have discovered that some trees can, with one single leaf, produce over 150V![5]

We can also try to explain how Max's powers work, based on a comparison with the eels. First, let's recall the fact that electricity is what makes our body work: the brain sends electric shocks (like orders basically) through the nervous system. Now, the eel behaves very similarly, except it can do something we can't: concentrate the electricity to one single spot. Then, Max would theoretically behave like an eel, by shifting the electricity coursing through his body in, say, his hands. But then, on a more morbid note, we can ask ourselves, if he could potentially direct the electricity from a distance. Is this just something that his brain does? (like an eel) Or could he absorb the electricity in someone else?[6]

1.4 WHERE HAVE ALL THE COWBOYS GONE

At the hospital, after the blackout, Kyle tells Liz that they are

> *"running a study on angiogenesis"*

upstairs. This is strongly related to Liz's field[7]:

DEFINITION 1.4.1: ANGIONESIS

Angionesis is the formation of new blood vessels. It plays a critical role in cancer research as scientists developed "angiogenesis inhibitors", that can block the growth of a tumor (since a tumor needs blood to grow) [10].

After checking Victor (the kid in the hospital), Kyle says that they have

> *"lost transcutaneous pacing."*

Another complicated scientific term.

[4] And now I can't stop thinking about Max hanging out with an eel...

[5] I am going off topic, but as I said above, I am really fascinated by cool science facts, and I will happily elaborate in another post, but for now, back to aliens!

[6] This is just me hypothesising, since the show barely explains how the alien powers work, or what are their limits. However, I do find the physics behind the powers to be extremely interesting, and therefore, in future chapters of that same pdf, you will find a longer explanation of alien powers, based on my personal theories.

[7] Again, more on Liz's research field later on.

DEFINITION 1.4.2: TRANSCUTANEOUS PACING

A **transcutaneous pacing** is a way of pacing a patient's heart during an emergency and stabilising the patient until further care can be admonished, by delivering short electric waves through the patient's body [11].

In this case, since the town had suffered a blackout, the hospital couldn't provide a pacemaker, hence why they used a transcutaneous pacing. But even then, some power is needed, which is why they lost it.

1.5 DON'T SPEAK

The only piece of science in this episode involves Liz detailing a bit more about her research: stem cell research, and vessel regrowth.

> *"But the work that you're doing with vessel regrowth is amazing. I mean, a few years ago, people never could have imagined this kind of progress in stem cell research. I mean, it's borderline science fiction. Look, this kind of science, healing what others believe to be irreparable, is important to me. It's all I've ever wanted to do."*

DEFINITION 1.5.1: STEM CELLS

Stem cells are the body's base cells, all the other cells are generated from the stem cells [12].

This research is a rather controversial field (as Liz mentions), namely since it involves the destruction of human embryos, and that the subject of when life begins is highly controversial – like the debate on abortion. We do see more on the controversy in 1x10, with the protesters outside the hospital, and later in 2x01 when Liz talks about regenerative medicine.

1.7 I SAW THE SIGN

[TW: AUTOPSY, DISSECTION]

Harlan Manes does the autopsy of an alien, and the scientist Everett mentions that he is

> *"Commencing the midline laparotomy."*

DEFINITION 1.7.1: LAPAROTOMY

A **laparotomy** is a cut that is made on the abdomen of a person (whether for autopsies, dissections, or surgeries) [13].

Midline here simply means that the cut is in the middle of the abdomen. Usually the cut is made deep and long enough to grant access to the internal organs.

1.8 BARELY BREATHING

Liz talks about the serum she created.

> *"I was working on this serum, a way to strip Isobel of her powers. I figured it might be a way to help people. It eliminates the electromagnetic charge around Max's squamous epithelial cells. So I hypothesize that the charge is connected to their powers. So, for all intents and purposes, it could render the aliens powerless. Human. But, I mean, it could do a lot worse, too. I don't know how much of their function is connected to this charge, so... I don't exactly have an alien mouse colony to test."*[8]

DEFINITION 1.8.1: EPITHELIUM

The **epithelium** is a type of tissue (along with connective tissue, muscular tissue and nervous tissue). It covers the outside of the organs and of the blood vessels through the body. There are three types of epithelial cells, but I will only go into the squamous type, as it is the one mentioned here [14].

DEFINITION 1.8.2: SQUAMOUS CELLS

Squamous cells can be found on the top layer of skin, in the lungs and around the heart and the blood vessels, and are constantly shed as new ones form. These cells filter what passes through the lungs, the heart and the vessels, and secrete a lubrification substance to help with the flow. It is also the most common type of cell found in urine, as the lubricant it secretes travels through the body and ends up in the bladder (though a too large amount could indicate a urine infection or certain types of cancer) [14].

Liz's sentence her confuses me, and sadly I do not know any biology students to help me out. So all I have to rely on are the articles I found online and in the university library. But from what I've read, the squamous epithelial cells do not appear to be carrying an electrical charge (or at least not one that would be significant enough to have massive consequence as it is eliminated).

However, there is a kind of epithelial cells (not the squamous type) that does carry an impressive charge of ions, namely the cells that help to our hearing (long story short, we can hear because we have little hairs in our ears that move with the air vibrations of sound, and those hairs send an electric charge to the brain, indicating that we are hearing something). Another kind of epithelial cell that carries a charge is the one that can be found in the retina (once again, by sending info to the brain) [15, 16, 17].

So unless the serum makes aliens somehow deaf or blind, I'm going to assume that this science fact is purely an alien one[9].

[8] I am keeping the lines about Liz's theory on how the aliens get their powers because I find it an interesting theory, but I am not going to really comment on it in further detail since this is of the realm of fiction. More aspects of the alien powers are analysed later, especially with the whole thing with frequencies.

[9] Please bear in mind I am a maths major, and sadly biology was never a field I was good at. All my knowledge comes from reading various articles in the little free time I have, so if this is incorrect, I sincerely apologise.

Liz then keeps on going with scientific terms:

> *"Okay, so we'll occlude the coronary artery of half the mice, reperfuse it..."*

DEFINITION 1.8.3: CORONARY ARTERY

To **occlude the coronary artery** means that we reduce or block the blood flow that goes through one of the *coronary arteries*.
The **coronary arteries** are the arteries that bring oxygenated blood to the heart [18].

Occluding the coronary artery then helps to stop a potential heart attach from happening. This can be done by inserting another vessel to bypass the narrow artery and improve the blood flow to the heart.

We then see Kyle examining Isobel's cells:

> *"The serum is attacking her cytoplasm. Her cells are rotting."*

I think Kyle explains pretty nicely what is going with her cells, but just so that everyone is on the same boat:

DEFINITION 1.8.4: CYTOPLASM

The **cytoplasm** is the gelatinous liquid found inside of a cell [19].

Hence, if the serum attacks the cells from the inside, the cells basically end up like an apple being eaten by a worm from the inside out (not an ideal condition).

The next bit is real-life science applied to aliens. We indeed have that Liz is working with an alien pod, which leaves little to no place for realism. She says

> *"I combined synthetic exothermic enzymes and melted down my jewelry to create a solution that should manipulate the properties of the membrane."*

Considering she is talking about an alien pod, the properties of the membrane and the silver factor are all clearly science fiction. But, to focus on the science here, what does "synthetic exothermic enzymes" mean?

DEFINITION 1.8.5: ENZYME

An **enzyme** is a *protein* that acts as a *catalyst* [20].

DEFINITION 1.8.6: PROTEIN

A **protein** is a highly complex substance present in all living organisms. Proteins of one species differ from the proteins of another species, and they also differ between organs. For instance, a brain protein is different to a muscle protein [21].

DEFINITION 1.8.7: CATALYST

A **catalyst** is a component that accelerates a chemical reaction [22].

DEFINITION 1.8.8: EXOTHERMIC

Exothermic is a term used in thermodynamics (the dynamics of energy, namely heat) that describes a reaction where the system involved releases energy to its surroundings, mostly under the form of heat [23].

In this case here, the system involved constitutes of the enzymes in whatever reaction they found themselves in, possibly the breaching of the pod. My guess here, is that the exothermic enzymes from melting the silver helps breach the pod by releasing heat during the chemical reaction. The heat then separates the cells from the pod, allowing entry.

1.10 I DON'T WANT TO MISS A THING

[CW: MATHS]

> *"Oh, you know. Just a fragment of a 70-year-old UFO. Your dad left it for me in the cabin. Some of the symbols in these letters match the symbols on this glass. I wonder if it correlates to a specific mathematical cipher. I'll start with the Fibonacci sequence, and then maybe Shor's algorithm."*

Alex is trying to break the code left by Jim to Kyle. Just like he said, let's start with the Fibonacci sequence.

DEFINITION 1.10.1: FIBONACCI NUMBERS

The **Fibonacci numbers** (denoted F_n) form a particular sequence where each number is the sum of the previous two. The first term changes depending on the author[a], but if we start at 0 and 1 we get [24]:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, \ldots$$

[a] Some omit the first two terms and instead start at 1 and 2, others at 1 and 1.

In Mathematics, these numbers often appear unexpectedly (trust me, when you least expect it, you end up with a Fibonacci sequence). The nth term of the Fibonacci sequence can be expressed in terms of n and of the golden ratio.

DEFINITION 1.10.2: GOLDEN RATIO

The **golden ratio** is a really cool maths tool, when two quantities have the same ratio as their sum and the largest term. We say that a and b have a golden ratio if [25]

$$\frac{a+b}{a} = \frac{a}{b}, \quad \text{when} \quad a > b > 0.$$

But enough of me going crazy over maths, here Alex is referencing the Fibonacci search sequence.

DEFINITION 1.10.3: FIBONACCI SEARCH SEQUENCE

The **Fibonacci search sequence** is a method of searching a searched array using a divide and conquer algorithm [26, 27].

Basically, say we are looking for a x term in the array. We divide the array in 2 and search both halves for x. Then, we divide the half with x into 2 again and search for x. We repeat until we have isolated the x (i.e. we can restrict it to an open neighbourhood with radius arbitrarily small).

In this case, I assume Alex would enter the message Jim left Kyle as entries of the array, then would try to find a "letter" which can be recognised and isolated, and would apply the Fibonacci search sequence until he found a pattern for each symbol, leading to a potential translation. Said like this, it does seem tedious, but in fact, if we start with a knows "letter" or "word" (word would be preferred, less computation time, and a smaller error), here the word being Magoo. As Kyle said, his dad was not a cryptographer, in the case that he was, Alex still would have been able to use the Fibonacci search sequence, but it would have required more lines of code to check for errors, and potentially another search sequence to help.

The second algorithm Alex mentions is **Shor's algorithm**. This algorithm is used to find the prime factors of an integers, using quantum computers. We can also apply it to crack public-key cryptographies (which require two keys, a public one that can be known by others, and a private one which is only known by the owner) [28, 29]. In this case, Alex would need the decryption key (the public one, which allows to decrypt the crypted message), or at least once again a known word to search through. This algorithm is more complex that the Fibonacci sequence, as it is based on quantum fundamentals, which I haven't yet properly studied, hence why I didn't go as much on a rant as for Fibonacci).

SEASON 2

2.1 STAY (I MISSED YOU)

[TW: AUTOPSY, DISSECTION]

During Noah's autopsy in Michael's bunker, Liz asks

"It could be some sort of organic phosphorus?"

DEFINITION 2.1.1: PHOSPHORUS

Phosphorus is a chemical element that can be found under two forms: white and red[a]. White phosphorus has the particularity that it emits a faint glow when it is exposed to oxygen[b][30].

[a] There are other forms, but there are a lot less common on Earth, and I haven't been on Oasis to check which form are present there.

[b] Where the term phosphorescence comes from.

We mostly (approximately 85%) find phosphorus in bones and teeth in the human body. White phosphorus is also found in fine china (as bone-ash) or in military-used incendiary bombs and smoke bombs. It causes burns and could have a psychological impact on the people affected, so it's use is very controlled.

There are two types of phosphorus: organic and inorganic. Inorganic phosphorus is made from a mixture of various minerals assembled together, while organic phosphorus is the one we find in plants and animals. The reason for inorganic phosphorus is that the extraction of organic phosphorus is expensive and mostly considered illegal (in big quantities at least).

What Liz is talking about here is that she is theorising that Noah's organs could contain traces of organic phosphorus, which isn't the case for humans, as she doesn't seem to be talking about Noah's bones.

Michael's response here goes back to what I talked about for 1x01, when he talks about Phenyl-2-Propanone (see Definition 1.1.1): "It's an unknown chemical compound. A distant derivative of phenyl-2-propanone. I could never figure out the 21st compound. But apparently it's the key to their nanotech and their biology [...] if it's organic, the tech itself could have neural activity."

We already know thanks to Alex and Michael's interaction in 1x01, that there is an (unknown) component close to P2P present in the alien's biology and in their tech. Liz and Michael are discussing that alien organs are "part tech" (as Liz says), since they are made from the same components are the alien tech Michael found from the pods and the ship.

We can then conclude that the alien tech and the alien DNA are closely linked. In both, a chemical close to P2P can be found, where the 21st component differs[10], as confirmed by the added information we have on the original script.

Note that neural activity modulates the connections between neurons. This furthermore tells us that the alien tech shares neural activity, and thus, is a living organism. Which would then explain the way the pieces of the ship want to be together, or the way they react to touch.

2.2 LADIES AND GENTLEMEN WE ARE FLOATING IN SPACE

As Liz draws blood from Isobel's arm, she says

> *"Human tissue can obviously regenerate from stem cells. With the right methodology, I could use your blood to make adjustments for alien physiology. I have to monitor exactly when cell degradation begins, down to the second."*

As I said in the first part, all of the body's other cells are generated from the stem cells (Definition 1.5.1), hence why human tissue can be regenerated from stem cells. What Liz and Michael are attempting here is to replace the "broken" parts of Noah's heart with "new" ones (hence why Liz is taking Isobel's blood I'm guessing).

I am not a medical major, but my research would lead me to believe that this is somewhat possible. And by somewhat, I mean that to an extent, scientist can indeed "mend" broken parts in a human (e.g. transplants). But there are limitations to such a procedure, that don't seem to exist in Liz's lab (or in the show in general) [31, 32, 33].

2.3 GOOD MOTHER

Liz explains her theory about bringing Max back to life to Jenna. She says

> *"Three years ago, I hypothesized that if I introduced a rare protein to destroyed stem cells, they'd regenerate. And I was right. My team in Denver brought dead cells back to life. Rat cells, but, I mean, still, the applications are immeasurable... until our study got shut down. They said it was for ethics reasons, but I think it's because it threatened big pharma. Then a few weeks ago, Kyle found that the pods contain a sort of alien cousin to my regenerative protein. When Max healed Rosa, his electric charge amplified the process. If we can replicate that, then, we can accelerate his recovery, so, I am testing out pig hearts to see..."*

I am definitely less knowledgeable than Liz when it comes to biology. After diving deep into stem cell research, reading many many articles about a field that is the furthest away from my major, I discovered that

[10] Considering even the alien expert doesn't know what it is, it's useless for me to to ever try to guess.

stem cells can indeed be regrown.

From what I've read, stem cells can divide and forms daughter cells, which can in term become stem cells again. This process is called self-renewal, and considering it happens on its own, I'm assuming that's not what Liz's research is about. No other cell in the body has the natural ability of generating new cell types [34].

There is however a way to repair damaged tissue or organs using embryonic stem cells. These new stem cells are taken from an embryo (given by a donor) that is three to five days old. This is the most controversial part of the research, as I said in the first part, with the debate on abortion [35]. In more recent studies (which could be the study that Liz took part of), scientists have discovered that there are also adult stem cells, i.e. stem cells from bone or fat of a (donor) adult.

These cells have more limited regenerative abilities compared to the embryos, but scientists have found a way to alter them (a method called genetic reprogramming, where scientists activate different cell types to modify the gene network. In simpler terms, this means that we can convert cell types in different cell types thus changing the properties.) By altering the adult stem cells, researchers can reprogram them to act in a similar way to embryo cells [36]. Even more recent studies show that it is possible to cure some diseases using the patient's own adult stem cells [37].

What Liz says here allowed me to look deeper in what her research could be about. I suppose that it is on transforming the adult stem cells into embryo cells, as this is done by introducing a *protein* (Definition 1.8.6) to the cells in order to genetically reprogram them. Please note that we are not given an exact explanation of her research, so this is all hypotheses.

2.4 WHAT IF GOD WAS ONE OF US

[CW: VERY FICTIVE SCIENCE]

While theorising about alien heart transplant, Liz says:

> *"Once Max is healthy, we could use this genome machine to target cellular apoptosis. I mean, we could craft polymerase sequencing in human DNA. We don't have to stop. We have no boards, no restrictions."*

DEFINITION 2.4.1: GENOME MACHINE

The **genome machine**, also known as DNA sequencer, is an instrument that automates DNA sequencing.
DNA sequencing is the process used to determine the order of nucleotides in DNA.
In biology, **sequencing** means to determine the primary structure of an unbranched biopolymer [38,

39].

DEFINITION 2.4.2: GENOME

The **genome** is the complete set of genetic information in an organism [40].

DEFINITION 2.4.3: NUCLEOTIDES

Nucleotides are organic molecules which are at the base of all hereditary characteristics [41].

When doing DNA sequencing to obtain the genome information, we basically read off all the nucleotides, or in other words, determine all the chromosomes that were inherited from our parents.

DEFINITION 2.4.4: CELLULAR APOPTOSIS

Cellular apoptosis is a programmed cell death in a highly regulated and controlled environment[a]. This happens naturally in the body. On average, a human loses between 50 and 70 billions cell each day due to apoptosis. Cellular apoptosis can help to eliminate the cells that have been damaged beyond repair (e.g. in cancer research) [42].

[a] i.e. unlike an injury or an illness that would cause the destruction of a cell.

DEFINITION 2.4.5: POLYMERASE

A **polymerase** is an enzyme (Definition 1.8.5) that is used to assemble DNA molecules [43].

DEFINITION 2.4.6: BIOPOLYMERS

A **biopolymer** is a material consisting of very larges molecules produced by the cells of living organisms. We find biopolymers in DNA (particularly, we call those biopolymers **polynucleotides**) which are linked together to form a helix [44].

As said in the content warning for this episode, this is purely fictive. To give you an idea, I understand the words separately, but not when they are put all together to form this sentence. Hence, *I* would tend to conclude it's not real. But let's translate exactly what Liz wants to do, because it's very nice and all to list you the definitions of the complicated terms, the point here is to actually explain it.

What Liz is suggesting they do here is purely fictional as far as my research went. I did not find anything that would indicate its possibility. Basically, she wants to use Max's DNA to choose which cells would be eliminated and which ones wouldn't. That in itself – if we ignore the fact that Max is an alien – isn't possible. Scientists cannot be that precise when eradicating a cell group. In such a procedure, the surrounding cells might be affected, which would lead to complications instead of a cure [45]. We can see more of that when Liz injects Steph (and more on the ethical question surrounding cell research as pointed out by Kyle), but I might not dive into it as it's mostly science fiction[11].

She also wants to be able to determine the primary structure of the polymerase in someone's DNA, which isn't possible[12], as the polynucleotides are linked, and therefore branched. If scientists were to debranch the polynucleotides, their molecular composition might change as some reactions might not occur. Hence, what Liz wants to do with Max's DNA is fictional, but is still a breakthrough nonetheless. Once again, we might ask ourselves the question of morality. This question often shows up in cell research (as already said, it's a very controversial field). I won't dive into such questions, however, as I am no biologist.

2.6 SEX AND CANDY

Kyle sees Steph in the room above the surgery room and he asks her

> *"You're here for the surgical separation of craniopagus twins?"*

DEFINITION 2.6.1: CRANIOPAGUS TWINS

Craniopagus twins are conjoined twins that are attached to one another at the cranium. Their brains are usually separate but they may share some brain tissue. It is an extremely rare condition, with approximately 1 in 2.5 million births, and the rate of survival after the first 30 days is very low (15 out of 50 per year, approximately) [46].

The separation surgery was first designed by computer-aided model design, then by the construction of a 3D printed model. The idea of this surgery is to passively create a separation between the twins over the course of several weeks, starting when they are 3 months old.

Generally, this distraction is performed over the course of 3 weeks at a rate of 2mm per day. At 7 months old, tissue expanders are placed circumferentially around the scalp and forehead, at the place of fusion, to create more skin coverage when it is time for the separation and the closure of the scalp.

Finally, there is a two stage surgery (one of which Kyle and Steph are attending) where in each stage surgeons separate one of the hemispheres. Complications can occur in either stage, and the time between each stage is dependent on the circumstances[13].

2.9 THE DINER

Liz is trying to understand what makes the earthly diseases so she could theoretically cure all humans diseases:

[11] At least for now, who knows what scientists discover in the future?
[12] Again, as far as I could see in my research, please correct me if I'm wrong.
[13] For information, the surgery described above is one of the methods that is most commonly used, but not the only one. Each set of conjoined twins is different, and doctors act accordingly.

> *"Hypothesis. The specimens display resistance to earthly infectious diseases due to alien hereditary immunity. Hypothesis. Pooled immunoglobulin harvested from an alien donor can be transferred to a human recipient. Human subjects exposed to specific sequencing of alien DNA will develop immunity to corresponding diseases."*

DEFINITION 2.9.1: POOLED IMMUNOGLOBULIN

Pooled means to accumulate in a same place.
An **immunoglobulin** is a protein (Definition 1.8.6) used by the immune system to identify and neutralise foreign objects such as bacteria or viruses [47].

Once again, this is mostly fictional, with hints of reality in it. By that, I mean that Liz is basically trying to create a vaccine, using alien DNA to combat viruses instead of a microdose of virus itself. This is where the fiction is: immunoglobulin transfer exists only (as far as my reading went) between a pregnant person and a fetus. However, the idea of protecting ourselves against a virus already exists: the vaccine.

Hence, in conclusion, what Liz says is 99.99% alien science, and 0.01% real-life science. Vaccines already exist.

2.10 AMERICAN WOMAN

[TW: ANIMAL TESTING, DRUGS, AND BUTYRICOL (AKA MY VILLAIN ORIGIN STORY)]

Liz asks Diego[14]

> *"Do you know what butyricol is? Worth a shot. It's this chemical I found in my friend's tox screen. I have never heard of it."*

Please note that **butyricol** (Definition 2.10.5) is something the writers completely made up from the show, but they did take inspiration from the real-world[15]. We know it's a memory eraser, and there are indeed drugs that exist that can do that (they are used in a controlled environment, usually to treat severe trauma). Once again, there is a huge ethical debate surrounding the use of such drugs (the question of where to draw a line comes up quite often).

To be clear, no magic, memory erasing drug has been developed or has hit the market (yet), and all the scenarios in *Roswell, New Mexico* surrounding butyricol are purely fictional. But, scientists have made progress in that direction, namely with one drug:

[14] I hate Diego. All his scenes gave me too much research, and too many headaches.
[15] At least, I hope they did. Would be a mad funny coincidence.

DEFINITION 2.10.1: PROPANOLOL

Propanolol is a drug that is currently under investigation for being a potential treatment to PTSD. Doctors sometimes prescribe it for high blood pressure, heart rhythm disorders, migraines, important tremors, excessive feelings of anxiety and an overactive thyroid gland[a] [48].

[a] At least, they can prescribe it Europe, no idea about the rest of the world, and since the subject at hand is memory, I won't dive in too deep into propanolol.

Now, the memory blocking part of propanolol. Propanolol inhibits the actions of norepinephrine,

DEFINITION 2.10.2: NOREPINEPHRINE

Norepinephrine is an organic chemical in both the brain and the body that functions as hormone and *neurotransmitter* [49].

DEFINITION 2.10.3: NEUROTRANSMITTER

A **neurotransmitter** is a signaling molecule that is secreted by a nerve cell to affect another cell [50].

One of the particularities of norepinephrine is that it enhances memory consolidation, which is the process of imprinting a memory in the hippocampus (in broad terms). A small study showed that patients who received a small dose of propanolol directly after a traumatic event experienced fewer stress related symptoms and a lower rate of PTSD than patients who hadn't.

Since memories are reconsolidated in mere hours after they are experienced, scientists discovered that propanolol could also help alleviate the emotional impact of memories that had already been formed. This means that propanolol could be used in the treatment of specific phobias.

Tw drug use, alcoholism (response to PTSD and trauma)
As I said earlier, there is a strong debate around the use of such drugs, namely in the treatment of PTSD. This debate is formed by legal and ethical questions, specifically on altering the memory that can be considered evidence during an investigation, or modifying the behavioral response to a past event, which could lead to a more severe response. However, the freedom of choice as been argued by scientists, especially given the fact that many who suffer from PTSD tend to use drugs and alcohol for this exact purpose [51].

We can also dig into butyricol by looking at butyric acid[16]:

DEFINITION 2.10.4: BUTYRIC ACID

Butyric acid, also known as butanoic acid is a chemical mostly found in various paints. It can also be used in mediating the development of an addiction, by regulating the long-term memory [52].

[16] Fun fact, butyric is derived from the Greek meaning butter.

Tw drugs, addiction, testing on animals

Scientists have discovered a link between butyric acid and memory, however it isn't even close to erasing someone's memory. It can only regulate whether the addiction reaches the long term memory and make sure the addiction cannot develop further. However, it is still only used in animals as a test[17] [53].

To conclude, I hate butyricol[18]. I wouldn't even consider it pure alien science, as it has nothing to do with our dear pod squad. It was purely made up for plot purposes. And I'm giving the writers a sad face emoji for choosing a word that doesn't have any etymological relation to memory, or drugs that could somewhat erase the memory. However, to keep consistent with the rest of the essay, I am going to give a definition of butyricol that isn't totally sarcastic.

DEFINITION 2.10.5: BUTYRICOL

Butyricol[a] is a memory erasing drug used purely in the TV show *Roswell, New Mexico*. It is presented as a liquid substance that is injected via a syringe. It's effects are almost immediate, and the amount of memories lost seems to be proportional to the dosage administered. It appears that the effects of butyricol cannot be reversed (yet).
It is theorised[b] that butyricol uses the toxins secreted by the diphteria bacteria (Definition 2.11.4) to interrupt the creation of protein (Definition 1.8.6) molecules, causing the interruption of communication between cells. Eventually, the lack of communication between cells will lead to amnesia.

[a] If anyone wants to add anything to this definition, feel free to reach out.
[b] See in the following section.

2.11 LINGER

[CW: A BUNCH OF MADE UP SCIENCE, MORE MENTIONS OF BUTYRICOL]

Flashback with Liz and Diego.

Diego: *"So, what are you doing with this worm behind my back?"*

Liz: *"Oh, administering a lethal dose of radiation to observe a particular neoblast. Don't be jealous."*

Liz: *"It's a regenerative master cell. The only one with this protein. I mean, this particular cell was able to multiply, diversify and reanimate my worm."*

Diego: *"Wait. So observing the master gets you the underlying mechanisms of tissue regeneration."*

Liz: *"Then I apply the mechanism to human tissue, and irreversible injury and degradation become distant memory."*

[17] So there is a link with memory, which could explain the etymology of butyricol, but doesn't seem to have any further comparisons.
[18] Actually, I don't. I think it's an interesting subject and all that. But the fact is, I lost many hours of sleep over it.

DEFINITION 2.11.1: NEOBLAST

A **neoblast** is a *non-differentiated cell* that can be found in planarians, which are a type of *flatworm* [54].
A **non-differentiated cell** is a cell that is still in the stem cell form, the body's base cells from which the others cells are differentiated [55].
A **flatworm** is a classification level in biology which regroups *bilaterian*, unsegmented (e.g. not like an earthworm), soft-bodied invertebrates (animals that do not have a vertebral column). Planarians are a type of plankton [56].
Bilaterians are animals with a left and a right side which are somewhat mirrors of each other (e.g. humans) [57].

Neoblasts can be selectively destroyed using X-ray, which lead scientists to study the comparison of organisms which lack stem cells with organisms that are **wild type worms** (some type of hermaphroditic worm of about 1nm length[19] that has no respiratory or circulatory systems). Scientists have compared planarians which were treated with a low dose of X-ray with specimens treated with a high dose. In those that had a low dose, they discovered that after some time, some radiotolerant neoblasts repopulated the planarians. In simpler terms, they microdosed the planarians with radiation so that they would build a resistance to such radiations [58].

So, I would tend to conclude that the science discussed here is somewhat accurate, but we can get deeper into the subject with Liz's second sentence, recopied here for clarity

> *"[The neoblast is] a regenerative master cell. The only one with this protein. [...] [It] was able to multiply, diversify and reanimate my worm."*

DEFINITION 2.11.2: REGENERATIVE MASTER CELL

A **regenerative master cell** is a stem cell which has the particularity that it is able to regenerate itself [59].

I am not entirely sure what Liz means by "the only one with this protein", but from the researches I've read on neoblasts, the subjects which had been administered a low dose of radiation developed new genes, which contained a particular set of proteins. These proteins are known to regulate the balance between the death and survival of the cell.

Thus, from the presence of these proteins[20], scientists were able to conclude that the the **cytoprotection** (i.e. protection of the cell by the hosting body with means of chemical compounds), the proliferation and the migration of the neoblasts are activated by a low dosage of X-ray [60].

[19] 1nm (nano-meter) = 0.000000001 m (meter).
[20] I am here assuming that Liz is talking about one of those proteins, but it may very well not be the case.

Hence, after administering a certain (unknown by us) dose of radiation to neoblasts, Liz developed a protein (also unknown to us) that showcased the multiplication, diversification, and reanimation of the worm. In other words, the cells were reanimated, which is one of the main parts of stem cell research.

In Liz's sentence here, the writers use a lot of vague words, so it does make sense from a scientific point of view, but it feels a lot far fetched. I'm guessing the vagueness of the terms, and the fact that most components are unknown to us is mostly to help make it more believable, and less easy to get wrong[21].

Later on in the episode, Diego[22] mentions once again butyricol:

> *"I bet butyricol uses inducible diphtheria toxins to stun the memory expression neurons into paralysis. See, if we can counteract that, we can get your friend her memories back."*[23]

As said above, butyricol (Definition 2.10.5) is fictive, and with propanolol (Definition 2.10.1), there is no way to retrieve memories (or at least as far as my research went). But let's decipher what Diego is talking about.

DEFINITION 2.11.3: INDUCIBLE

In biochemistry, **inducible** (when talking about enzymes (Definition 1.8.5)) means produced in response to the presence of an appropriate inducer [61].

I know, this is the vaguest definition ever, but bear with me for what follows.

DEFINITION 2.11.4: DIPHTERIA

Diphtheria is an infection cause by the bacterium called Corynebacterium diphthariae. The symptoms of diphtheria are pretty gruesome so I will spare you the details, but basically they include a strong fever (38°C or 100.4°F or above), chills, fatigue, blueish skin coloration, sore throat, headache, difficulty breathing, and the formation of dead tissue in the respiratory system[a].

Diphtheria toxin is a toxin secreted by the Corynebacterium diphthariae bacteria and causes diphtheria by entering the cell cytoplasm and inhibiting *protein synthesis* (process of creating protein molecules) [62].

[a] This is the gruesome thing I'm talking about, I wouldn't wish it on my worst enemy.

Scientists have discovered that inhibiting protein synthesis can lead to amnesia. I am not entirely sure I understood it all, but basically it all boils down to the communication between cells, and that if one cell stop

[21] The writers have understood the idea of academia very well: being as vague as possible to not get points deducted.

[22] Have I mentioned I hate the guy? He is giving me more headaches than academia!

[23] The words used here (other than butyricol, for obvious reasons), are all existing words, that do actually make sense in a sentence (round of applause please!)

creating protein molecules, the brain will eventually be touched, and stop retaining memories, or ditch some memories, depending on the severity of the inhibition [63, 64].

Hence, Diego is theorizing that butyricol uses the toxins secreted by the diptheria bacteria to stop protein synthesis and thus lead to memory loss.[24]

However, as I said above, it isn't possible to reverse this effect. It also seems that the memory loss induced by inhibition of protein synthesis is rather small compared to the effects we see in *Roswell, New Mexico* with butyricol, or what scientists observed with propanolol. Hence what Diego is suggesting they do here is once again, fiction.

As a response to Diego's idea about butyricol, Liz mentions two acronyms:

> *"Would it boost GABA in DLPFC?"*[25]

DEFINITION 2.11.5: GABA, DLPCF

GABA stands for Gamma-Aminobutyric Acid, and is a neurotransmitter (Definition 2.10.3) that blocks impulses between nerve cells in the brain. Low levels of GABA can be linked to anxiety, mood disorders, epilepsy, or chronic pain. Scientists suggest that GABA can boost mood or calm the nervous system (here, GABA taken as a supplement) [65].

DLPCF stands for dorsolateral prefrontal cortex, an area in the prefrontal cortex of the brain of humans or other primates. One of the main function of the DLPCF is the executive functions, namely working memory [66].

Hence, since scientists have already established a connection between mood disorders and memory loss, we can conclude that a boost of GABA in the DLPCF would indeed increase memory (or, more precisely, lower memory loss, as we can't exactly boost our memory per say). Note that this is the idea behind the fact that depression and anxiety can cause memory loss [67].

I won't elaborate further, because otherwise I'd just be repeating myself, but Diego mentions that Liz is thinking about working memory and not traumatic memory, which are indeed the two main differences between low level of GABA and propanolol (Definition 2.10.1).

We then see Diego interrupting Liz grinding eggshells, where he says:

> *"You're making calcium carbonate. Is this about regenerating worms?"*[26]

[24] This is what he means by "stun the memory expression neurons into paralysis", idk why they had to say it in so many words, to sound more fancy perhaps.

[25] The acronyms exist, but again, as it follows from butyricol, it's fictional science.

[26] Honestly, we are given very little information, and this scene is way too vague, so I'm gonna assume it's made up science.

DEFINITION 2.11.6: CALCIUM CARBONATE

Calcium carbonate is a chemical compound that is commonly found in rocks, and is the main ingredient in eggshells, shellfish skeletons and pearls [68].

Fun fact, traces of calcium carbonate have been found in Mars, indicating the past presence of water, which could have lead to living organisms!

Diego then asks Liz if she's trying to regenerate worms, and I wish the show had given us more information, since those two facts are very little to start on. I did found that there is a link between worms and calcium carbonate, namely that worms can help regenerate (i.e. recreate damaged parts of) minerals. However, whether or not this is a two way street was very hard to find, and little existence showcases the fact that worms can regenerate from minerals.

There may be more behind all of this, but sadly nothing in that scene constitutes a basis strong enough for me to start my research on. Also, I am doubtful as to the plot related importance of that scene, since we do not see any more of the calcium carbonate.

2.13 MR JONES

At the hospital, after Maria tells Michael that Liz has been coming to inject her with a serum, he responds with:

> *"Ooh, the synthetic nucleotide excision repair genomogenate? Mm-hmm. We're lucky. You're only part alien, otherwise there wouldn't have been enough left of you for her to save."*

DEFINITION 2.13.1: EXCISION

Excision (in surgical terms) means to partially remove an organ, tissue or bone from a body. In biology, this word has a similar meaning, but applied to molecules, i.e. it is the removal of a section of DNA that is damaged due to a mutation or incorrect replication [69].

I already mentioned nucleotides in the analysis of 2x04 (Definition 2.4.6), but basically what Liz is doing by injecting Maria of a serum, is that she is repairing the branches of Maria's DNA which were damaged by the bomb. This is again completely alien science.

The word '**genomogenate**' is completely made up, and I don't really understand why. I mean they probably could've used an actual word (like the word genome). Even if it's something that is unique to aliens, Liz is the one who probably did the naming, and I'm assuming it would make her life greatly easier to not invent new words only few people are going to use.[27] However fictional this word may be, it seems to have its etymology in the word 'genome' definition 2.4.2

[27] Unless it's just because I'm a lazy mathematician.

By the context of Michael's sentence, I am tempted to assume that genomogenate probably designates the complete set of alien genetic information, but once again, why make up a completely new word since genome already applies to all living organisms. Unless aliens are somehow not living organisms, which I strongly doubt.

SEASON 3

3.1 HANDS

Heath[28] and Liz are at Genoryx, and Heath makes the following pun:

> *"Oh, please, Dr. Roswell, don't cut me. Cytosine promises he will never hook up with adenine ever again."*

DEFINITION 3.1.1: CYTOSINE, ADENINE

Cytosine and **adenine** are two of the four chemical bases of DNA (the other two are guanine and thymine). Adenine and cytosine cannot pair together (indeed, adenine pairs with thymine and cytosine with guanine)[70].

However, in a theoretical setting, a bond formed of hydrogen could be made between cytosine and adenine, which would be an isomorphic bond (meaning that the structures of the adenine and the cytosine would be preserved). Such a bond could lead to mutations in the DNA.

Judging by Heath's joke, I'm assuming that Liz and Heath are trying to find a way to create such mutations (namely for regeneration, as it is the subject of Liz's research), but without a cytosine-adenine pair (as that would lead to mutations that would be unwanted).

Please note that I am not a stem cell researcher, and the only article I could find on cytosine-adenine bonds was written in 1986 with way too many scientific terms I had never heard of before. Also, I am not a writer in the show, and I do not read minds.

We then see a *RNA* on Liz's screen:

DEFINITION 3.1.2: RNA

RNA stands for Ribonucleic acid, and is a molecule similar to DNA, which unlike DNA is single stranded. Here again, cytosine and adenine do not pair (however, this time, adenine's complement is uracil, whereas cytosine's pair is still guanine) [71].

Kyle arrives at Genoryx, and Liz tells him: "[I need] my antibodies from Noah's choroid plexus." We can already start to conclude that this is science applied to aliens.

[28] I like Heath. Heath talks about actual science and makes science puns. Heath does not give me headaches.

DEFINITION 3.1.3: PLEXUS

A **plexus** is a network of vessels or nerves. The **choroid plexus** is a plexus of cells that comes from each of the ventricles of the brain. The brain has four ventricles which are connected to each other, and the fourth ventricle is directly connected to the spinal cord[72].
The choroid plexus produces the **cerebrospinal fluid** (a clear colourless body fluid that surrounds the brain and the spinal cord) of the central nervous system [73].

Cerebrospinal fluid also carries a few antibodies (namely to protect the brain and the nervous system in the spine), which is what I believe Liz needs in her research.

Liz and Heath break into their boss' office, and Heath says:

> *"Hey, you know what? Remove a plasmid from a bacteria, I'm your guy. I'm not Ethan Hunt. I still have student loans."*[29]

DEFINITION 3.1.4: PLASMID

A **plasmid** is a molecule that is found in the DNA of bacteria (but not in chromosomes, hence we call it an extrachromosomal DNA molecule), that can replicate itself independently from the surrounding cells [74].

The **isolation**[a] of plasmid in bacteria is a technique in molecular biology that is used for cloning, DNA sequencing (Definition 2.4.1), **transfection** (basically the transfer of cells), and gene therapy.

[a] Or as Heath says here, removal, though we don't actually remove it, just isolate it from the other cells.

All of the aforementioned subjects are what Liz and Heath are studying to save their aliens (respectively Maria and Dallas).

3.3 BLACK HOLE SUN

Liz and Kyle enter Liz's old lab, and Liz says:

> *"My coworker at Genoryx reminded me that bacterial endospores can withstand even the most extreme environmental conditions. And he did it with rum."*[30]

[29] See, this is why I like Heath. Real-life science, once again. And I don't have to spend hours copy-pasting DOIs into sci-hub just to figure out what he's talking about (*looking at Diego*)

[30] Liz is stating an actual real-life science fact (I'm so proud of her)

DEFINITION 3.3.1: BACTERIAL ENDOSPORE

A **bacterial endospore** is a simplified formed of a bacteria that consists of the DNA genome (Definition 2.4.2), a small amount of cytoplasm (Definition 1.8.4), and a coating which allows the bacterial endospore to resist to heat, radiation, and many more types of harsh conditions [75].

Here, the harsh conditions are the fire from when Max burnt down Liz's lab. This then just means that part of Noah's DNA have survived the explosion.

3.5 KILLING ME SOFLTY WITH HIS SONG

The paramedics are working on Liz, who passed out after Max healed Kyle, and one of them says:

> *"Systolic BP's 170 and rising. Pushing 100 milligrams of lidocaine."*[31]

DEFINITION 3.5.1: SYSTOLIC BP

The **systolic BP** (blood pressure) is the pressure injected into the arteries when the heart beats.
Lidocaine is a local anaesthetic that is also used to treat ventricular tachycardia (fast heart rate) [76].

Normal systolic pressure is around 120 mmHg[32] or lower. Here, Liz having a pressure of 170 is indeed worrying.

When Liz and Heath are driving to Roswell, he tells her:

> *"You are simultaneously presenting symptoms of vascular degeneration and African trypanosomiasis."*

This is a real-life scientific notion, but I am struggling to see it's relevance.

DEFINITION 3.5.2: VASCULAR DEGENERATION

Vascular degeneration means that less blood flows through the body, which can lead to some functions to shut down (like the brain or the heart). Vascular degeneration can lead to neurodegenerative conditions, such as Alzheimer's or dementia [77].

DEFINITION 3.5.3: AFRICAN TRYPANOSOMIASIS

African trypanosomiasis is a parasitic infection, transmitted through insects, that can affect humans and animals. The first stage of the disease is characterised by headaches, fevers, pain in the joints, and itchiness. The symptoms can start appearing one to three weeks after the initial bite.
The second stage is characterised by confusion, poor coordination, trouble sleeping and numbness,

[31] I would assume real life science, but everything medical is not my cup of tea so...
[32] millimetre of mercury, 1mmHg = 133Pa (Pascals) in SI units.

and the symptoms can appear weeks to months after the first stage. In the second stage, the parasite enters the nervous system, which can cause long-lasting effects. Without treatment, this disease can be fatal [78, 79].

I haven't had time to rewatch season 3 (or much of the show really), so I've mostly been relying on the transcripts to figure out the science. So I don't really remember the symptoms that Liz is experiencing (or at least what we see on screen), but I would assume they are similar to those of the African trypanosomiasis, otherwise why would the writers mention this disease specifically?

3.6 BITTER SWEET SYMPHONY

Alex and Eduardo Ramos are at Deep Sky following the bat attacks. He says to Alex

> *"Laparoscopic technology is invasive."*

Real-life science, quite straightforward[33].

DEFINITION 3.6.1: LAPAROSCOPY

A **laparoscopy** is an operation done in the abdomen or the pelvis using small incisions and a camera (called the laparoscope) [80].

Here, Alex didn't become a surgeon overnight, but he did insert a laparoscope into the Lockhart machine, in order to see its inside (hence how he found the alien symbols). A laparoscope is basically a small camera at the end of a long metal rod (exactly like what Alex used to see the inside of the machine). Indeed, laparoscopic technology is invasive, as we stick a metal rod inside an organism, that may or may not be happy with the intrusion.

3.7 GOODNIGHT ELIZABETH

[CW: A VERY LONG RANT ABOUT FREQUENCIES]

Liz tries to help Maria. She makes the following hypothesis:

> *"alien cells use hydrolyse component polymers to attach themselves to host cells."*

Considering the sentence literally starts with 'alien cells', I think it's safe to say it's alien science. But it's not entirely made up, as it consists of actual scientific facts.

[33] Ramos and Heath are my favourite secondary characters at this point.

DEFINITION 3.7.1: POLYMER

A **polymer** is a material consisting of very large molecules. Polymers have a wide range of properties and can be found in synthetic plastics (e.g. polystyrene) or in DNA and proteins (Definition 1.8.6) [81].

DEFINITION 3.7.2: HYDROLYSIS

A **hydrolysis** is a chemical reaction where a molecule of water breaks one or more chemical bonds. In the case of polymers, they can indeed be susceptible to degradation by hydrolysis. A polymer degradation is a change in the properties of the polymer, i.e. colour, shape, molecular weight, or its strength. [82]

So basically, here Liz is saying that alien cells use degraded polymers (said degradation is done via a molecule of water) to attach themselves to human cells. This is a reaction that can happen, and is not only due to the alien cells: polymers, once degraded, have their structure that is changed and can thus attach themselves to other cells (with the right combination of cells).

Note that this is roughly how virus enter a body, by attaching themselves to cells and developing. In the case of Maria, Liz does make the comparison with a virus, but also mentions that the cells are (for now) not reproducing. This is due to the fact that Maria's cells are fighting the alien ones, thus stopping the spread of the 'virus'.

Michael and Rosa are at the junkyard, talking about frequencies. But before going into details of what is being said, first I will go on a little rant about frequencies, *and how everything as a frequency* (which is kinda what Michael says).

I have nothing to add to that scene, it's already perfect as it is; both with Michael and Rosa sharing a scene, and bantering, but also from a scientific standpoint. Everything Michael says in that scene is 100% accurate, and also a subject that I personally find fascinating: every single matter in the universe emits a certain wavelength. Think about it, everything resonates, albeit at a different frequency, but resonates nonetheless.

DEFINITION 3.7.3: HERTZ

The Hertz (Hz) is the international unit to measure frequency, where 1 Hz is equal to 1 cycle/second [83].

The human body is said to have a fundamental frequency (Definition 3.7.4) of about 5 to 10 Hz (this difference is due to different methods used to test it in different researches), the frequency of Earth is around 7.8 Hz, the Sun at 126 Hz, the Moon at 210 Hz, and black holes of 250 Hz[34].

[34] Do I know those values by heart? Maybe? You can't prove anything.

DEFINITION 3.7.4: FUNDAMENTAL FREQUENCY

The **fundamental frequency** is the lowest frequency of a periodic *waveform*. The fundamental frequency is perceived as the loudest frequency [84].
The **waveform** of a signal is the shape of its graph as a function of time [85].

Let's look at the Earth for example (the example I am most knowledgeable on). The fundamental frequency of the Earth is of 7.83 Hz, but what does that mean exactly? In the 1950's, a physicist named Winfried Schumann mathematically predicted the global electromagnetic resonance that is generated from an electricity discharge from the Earth's surface.

This electrical current is excited when it comes into contact with the ionosphere (the part of Earth's atmosphere that contains the most ions, which are atoms or molecule that posses an electrical charge). Schumann basically discovered that the electricity that courses through the atmosphere oscillates at a period that ranges from about 0.1 seconds to 0.03s (0.1s being the fundamental period of Earth).

In reality, Schumann didn't really invent anything. There was actually a physicist before him, FitzGerald in the 1890's, who did quite some groundbreaking theories and calculations that were at the foundation for the discovery of Earth's fundamental frequency. Namely, the fact that some layers of the atmosphere could be conductors. But FitzGerald didn't really publish his findings, so no one really knew what he did and very little credit get's attributed to him.

Note that the frequency is 1/period, hence 7.83 Hz corresponds to 0.1 s, which is the period that FitzGerald calculated originally, 60 years before Schumann came into the game. I guess the lesson here, kids, is to always publish your findings, otherwise an entire study that originated from you might carry someone else's name.

Anyway, enough of me fangirling over science, and back to our alien show. As I said, everything Michael says is true, but what exactly is he saying?

> *"You don't just hear things. Your ears are detecting subatomic vibrational frequencies."*

Subatomic refers to particles that are smaller than or within an atom, such as protons, neutrons, electrons. So, basically, Rosa can hear the frequency of the vibration of those tiny particles which consistute everything around us.

> *"All matter in the universe emits signature sound wavelengths: rocks, uh, wind, fire, plants, even people."*

See rant above.

Rosa: "Wait, is that how I was able to find Maria in her memories?"

Michael: "Neuronal activity produces brain waves oscillating at a high beta frequency. Your noggin picked *up on those. That's how you found her."*

I don't think I need to repeat that everything has a frequency.

DEFINITION 3.7.5: HIGH BETA WAVES

Beta waves are high-frequency, low-amplitude brain waves that are commonly observed in an awaken state. **High beta waves** are waves that have a frequency of 18-40 Hz [86].

Hence Michael says here that neuronal activity has a frequency that ranges from 18 to 40 Hz. This is not quite true. High beta waves (when talking about neural activity) are associated with significant stress, anxiety, paranoia, high energy, and high arousal. This means that high beta waves are one of the wave types describing neural activity. As a matter of fact, there are five types of wave frequencies: delta (1-4 Hz), theta (4-8 Hz), alpha (8-12 Hz), beta (13-40 Hz), and gamma (40-150 Hz). Both beta and gamma frequencies have sub-classifications: low, mid-range, and high for beta; low and high for gamma. So really, high beta frequencies are not all there is to neuronal activity.

"But I wonder, can you detect resting frequency gradients?"

DEFINITION 3.7.6: FREQUENCY GRADIENT

A **frequency gradient** is when all the signals are put together, which gives a large signal at the start, as all the signals start at the same point, but becomes smaller as the signal diverge (since they don't all have the same frequency).

What Michael is asking here, is whether Rosa's powers only work with 'pulses', namely from when Michael hits the alien glass, or whether she can pick up the fundamental frequency of resting objects. Spoiler, the answer is yes, Rosa can pick up the fundamental frequency of resting objects, as we see with the sword.

This is super cool, but also how far can her powers go? Because, as I keep saying, everything has a fundamental frequency, that's kinda the point of the universe. My guess, is that there must be a limit to a certain frequency, or wavelength, that she just cannot hear. Just like us humans, that can hear from about 20 Hz to 20KHz, Rosa's is just also higher.

And yes, this means that us humans can actually hear the Sun. Granted, maybe not from where we are, as, you know, the sun is 148 million km away from us, and sound travels 340m/s. I'll spare you the details, but it would take 435 thousand years for the sound to travel. I strongly invite you to watch youtube videos on the sound of sun, it's so cool.

3.8 FREE YOUR MIND

[CW: FAKE CODE]

Liz talks in her voice recorder, theorizing about what is happening to Maria and how to solve it.

> *"The nanoblades will now be combined with the newly modified alien serum and infiltrate the Jones-Maria cell cluster for a cellular separation test."*

What Liz is descriving is a real thing (nanoblades) but in an alien context (Jones).

DEFINITION 3.8.1: NANOBLADE

A **nanoblade** is an assembly of proteins, with the presence of virus (namely the mouse leukemia virus) and virus-like particles. Note that nanoblades actually lack viral genome (Definition 2.4.2) (what transmits the virus and helps propagation), hence they are not-infectious and non-replicative. Nanoblades are mostly used in editing the primary cells of the genome [87].

Basically, nanoblades infiltrate a cell cluster and help separate a virus from those cells (roughly speaking). This is exactly what is happening here, with Liz trying to separate Jones' cells from Maria. However, we can also see that it doesn't work, which is also something that can happen with humans, as nanoblades do not work for separating all types of viruses form cells (and sometimes the host body rejects the presence of nanoblades).

We then have Alex's infamous code. So, as many may know, this code is not a real code, and is actually a toy program. JoCarthage on Tumblr gave a very good rundown of the code in her post, that I strongly encourage everyone to read!

I tried to approach it from a more Watsonian point of view (also because I am a lot better at algrebraic maths than at numerical maths). In theory, what Alex is doing is possible (though most definitely not the way he did), and I believe that he would need to import the data (here the Roswell security footage).

The only time I ever imported data was for an academic paper on analysing the data from the stock market and predict the change in the course[35]. I'm assuming Alex's code would require to be a lot more detailed than mine, as he needs the data to be imported in real time.

Another way to explain how 'simple' Alex's code is in regards to what he is actually doing is that we only see the end of it, and therefore all the complicated stuff it above. He is also currently explaining it to Michael, who I'm assuming doesn't know that much coding.

We then go back to Liz, and once more on frequencies:

[35] And the only thing I'll say is that I would never bet money on it.

> *"I found this research explaining how sound frequency can break plaque bonds in Alzheimer's patients."*

I think I have found the same research paper as Liz, considering I only found one after many hours of internet libraries of scientific journals. However, this research hasn't yet been extended to humans. But maybe, since Liz is an actual Doctor, and I'm not (yet) one, she has access to more experimental stuff.

Basically, the idea here is that sound flickering at 40Hz could restore connectivity to the brain and mobilize the immune system to fight against toxic proteins that cause Alzheimer's [88]. This is roughly what Liz and Rosa are doing with Maria: using sound to separate the cells.

3.9 TONES OF HOME

[CW: LONG RANTS ABOUT CIPHERS]

We first see on Liz's board the word *harmonies*. Don't worry, I will not start listing the different fundamental frequencies, but I do like that the show has been consistent with that fact, not just saying that frequencies could sever the connections between cells once, but actually giving the viewer a greater insight to it.

On a general matter, I do like that the science is a inherent part of the show, and that it isn't just there to sound fancy, but that there is some actual research behind it, but that really isn't the subject at hand here.

Kyle is in the medical room at Deep Sky, talking to Darby.

> *"I suggest that you have your team relook at the endoplasmic reticulum for traces of the paralytic."*

Finally some actual real-life science!

DEFINITION 3.9.1: ENDOPLASMIC RETICULUM, EUKARYOTE CELLS

The **endoplasmic reticulum** is the means of transportation of the *eukaryote cells* [89].
Eukaryote cells are cells that contain a distinguishable nucleus, which is were the genome (Definition 2.4.2) of the cell is contained [90].

Note that the endoplasmic reticulum transports and sorts protein into the different zones where it is needed. Hence, this is why Kyle suggests that Darby's team look into the endoplasmic reticulum, as it could have carried traces of the paralytic (the thing Kyle was injected with at Max's house), and would then still contain traces of it.

We finally have Alex talking about ciphers, which is precisely the scene that gave me the idea for this very long meta.

> *"Working theory is a take on the Vigenère cipher with a twist."*

Don't get me wrong, I enjoy researching biological terms. But there is a reason chose to study maths. And you are about to get an inside view on my obsession with mathematics.

DEFINITION 3.9.2: VIGENÈRE CIPHER

A **cipher** is an algorithm that perfomrs encryption or decryption using a series of well defined steps that can be followed as a procedure [91].
The **Vigenère cipher** is an encryption method that dates back to the 1550s, first described by Giovan Battista Bellaso, that uses a series of Caesar[a] ciphers [92, 93].
The **Caesar cipher** is one of the most widely known encryption techniques, and is indeed named after Julius Caesar, who used it in his private communications [94, 95].

[a] Funny, as I'm currently writing this on the ides of March.

Basically, the Caesar cipher works as follows: we first determine a fixed number, and we then shift all the letters down the alphabet by this number. For example, if I wanted to send "hello world" as an encrypted message to Julius Caesar, with a fixed number of 5, I would send "CZGGJ RJMGY". He would then need to have the decryption key (here, 5), would count down the alphabet, and would be able to decipher my message.

The Vigenère cipher follows the same idea, hence it is rather easy to understand and to implement, but is much harder to break. That is, until 1863, when Friedrich Kasiski was the first to ever publish a general method to decrypt Vigenère ciphers.

Also, fun fact, but Blaise de Vigenère did not in fact invent the Vigenère cipher, but he did describe it in his *autokey cipher* (Definition 3.9.3), which is a way to code and decode many encryptions. However, Vigenère published his paper on the autokey after Bellaso first mentioned what we now know as the Vigenère cipher. Basically, Vigenère stole all the fame (whether he did it in purpose or not I don't know, as the cipher got this name in the 19th century, once both men were long gone).

DEFINITION 3.9.3: AUTOKEY CIPHER

An **autokey cipher** is a cipher for which the key is generated from the text you want to code, using previous parts of the text to encrypt [96, 97].

Another fun fact, this cipher is famous for being "le chiffrage indéchiffrable", which is french for the indecipherable cipher. Well, shall be decipher it then? Actually, we won't, because that might take too much time, and this essay is already long enough as it is, I don't want to bore you more. I will however explain the cipher and give an example. As I said above, the Vigenère cipher is constituted of many Caesar ciphers in one, with each their own shift value. To encrypt it, we can use a *tabula recta* (see Figure 3.1). To let the receiver know what the shift value is (i.e. which alphabet to use out of the 26x26 table), the sender choses a keyword and repeats it until it matches the length of the plain text (the text to be sent).

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Figure 3.1. A **tabula recta**: a square 26x26 table of alphabets, where each row is shifted to the left by one [98].

Let's try an example. If I want to send "helloworld" (this is then what we call the plain text), and I use a keyword "CAESAR"[36], this means that the encoded message will be such that each letter will be the intersection of {h,e,l,l,o,w,o,r,l,d} and {c,a,e,s,a,r,c,a,e,s}, i.e. this gives: "JEPDONQVLV". There is also an algebraic way of describing such a cipher, namely using the modulo class of 26, and numbering the letters A-Z as 0-25, but I am going off subject here, so that might be for another day. A code in Python (made by me) showing the encryption of both the Caesar and Vigenère ciphers is provided in Appendix 4.12.

To go back to the Lockhart machine, what I'm guessing Alex means here is that the machine holds a message, and that it is stored in a way similar to the Vigenère cipher, with its own alphabet (alien), and probably using its own version of a tabula recta, otherwise that would be too easy. I'm assuming Alex would know how to break the Vigenère cipher, especially since even I know the theory behind it, and could probably figure it out given enough time and the resources, which Alex no doubt has.

Side note, but the breaking of the Vigenère cipher relies on repetitive patterns, and can be deciphered using the algebraic definition, patience, and lots of ink and paper (the key here is patience, but it did happen that I arrived at the end of a pen trying to decipher it, so I would suggest coming prepared).

[36] It's the ides of March for me as I'm writing this, so I might as well honour the guy you know?

Side side note, but I am very much like Michael here, I love when Alex talks all code-y. As I said, there is a reason why this episode specifically made me wanna dive into the science, hearing about Vigenère cipher may have made me lose my mind.

Side side side note, but when Michael said that solving a 75-year old mystery seems like a pretty good first date, I could not agree more! But that might be the nerd in me speaking…

3.10 ANGELS OF THE SILENCE

Liz and Heath mention the acronym CRISPR a few times this episode:

Liz: *"Heath's still working under the assumption that CRISPR will splice through binding proteins."*

Heath: *"Oh, it's okay. Liz won't mind if I borrow some of her CRISPR chemicals."*

Liz: *"Maybe I could use that to reason with him. Heath still thinks CRISPR is the answer. There is only one place that he would go."*

Liz: *"Bandelier University keeps CRISPR in their biolab."*

Liz: *"CRISPR isn't gonna get you all the way there."*

Heath: *"Jones thinks CRISPR has a part to play in his endgame. I can buy you some time."*

I actually also did mention this acronym before, only indirectly.

DEFINITION 3.10.1: CRISPR

CRISPR stands for 'clustered regularly interspaced short palindromic repeats', and is a family of DNA sequences found in the genome of bacteria [99].

CRISPR, just like nanoblades (Definition 3.8.1), infiltrate a cluster of cell, and separate the cells from a virus. Hence, here, Heath thinks that CRISPR could help stop whatever is happening to Dallas' brain.[37]

3.12 I AIN'T GOIN' OUT LIKE THAT

Liz and Heath are teaming up at Deep Sky to separate Max and Jones.

Heath: *"Huh. Binding receptors are multiplying. They're effectively stopping the peplomers in Jones' cells from reattaching to Max's DNA. Oh. But at quite the cost."*

As you probably guessed it, this is mostly all alien science. But some of it is real!

[37] Except it's alien, so it won't work, but points for effort!

DEFINITION 3.12.1: BINDING RECEPTOR

A **binding receptor** is a molecule that attaches itself to another cell and transmits a message (neurotransmitters (Definition 2.10.3) are binding receptors) [100].

DEFINITION 3.12.2: PEPLOMER

A **peplomer** is a protein from a virus, that has the particularity of being spiked and hence attaching itself to a host and spreading the virus [101].

The coronavirus is a peplomer, so it operates in a similar way to Jones' cells. We then have that Liz and Heath are applying real-life science to aliens. And it seems to be working.

After helping Kyle take out the shards out of Dallas' neck, Isobel says

> *"4-0 Monocryl sutures ready to roll."*

This is mostly real-science, as it's just medical terms.

DEFINITION 3.12.3: MONOCRYL

Monocryl is a synthetic surgical suture, i.e. a medical device used to hold body tissue together, to help wounds close after a surgery. 4-0 refers the size of the thread [102].

Heath ends the science in season three with another real-life science fact[38], with a little alien twist.

> *"Uh, U11, U12, U5. We have all of the pieces. We're just missing the mystery protein: UX. Man, it's got to be the key to repairing the DNA while making it inaccessible to Jones' tether, so, close, but no cigar."*

UX, the mystery protein, can be interpreted as a 'find x' problem.

U11, U12, and U5 are proteins found in the RNA (Definition 3.1.2). I would like to stress the fact that these proteins exist, and are not made up for the purposes of sounding more fancy.

Honestly, what these proteins do is rather confusing, and I don't want to make stuff up if I'm not at least a little bit sure of what I'm saying. All I can tell, however, is that these proteins are important for the 'slicing' of molecules, i.e. when we separate molecules from an invasive protein. Hence why they are mentioned here. I'm assuming this UX would act similarly, but as Heath says, it's purely alien, so really, there is no telling what this unknown UX could do.

[38] Told you, I like the guy for a reason.

SEASON 4

4.1 STEAL MY SUNSHINE

We start off the science in this season with Liz talking about the Leidenfrost effect.

> *"Based on week seven's quiz results, we're going to do a refresh on the Leidenfrost Effect."*

I am very proud and happy that this season begins with some real life science!

DEFINITION 4.1.1: LEIDENFROST EFFECT

The **Leidenfrost effect** is when a liquid, that is close to a surface significantly hotter than the liquid's boiling point, produces an insulating vapor that stops it from boiling rapidly. We can then notice that the drop of liquid will hover above the hot surface [103].

You can experiment with the Leidenfrost effect when cooking. Say you have a pan at roughly 190°C (which corresponds to the Leidenfrost point for water, and according to google this is 374°F) and you put some water on the pan. Since water boils at 100°C (=212°F), the surface of the pan is then sufficiently hot to observe the droplet of water hovering for a bit, and they will not evaporate as fast as if the pan had been cooler. I always sprinkle a bit of water on hot pans whenever I cook, ever since my physics teacher did this in class. It's also a great way to know when your stainless steal pan is hot enough for your e.g. scrambled eggs not to stick!.

Liz then continues by doing an experiment in front of her GED class[39]. She says the following, and then sets fire to the mixture,

> *"All right. This is potassium nitrate. This is common table sugar. Apart... nada. But together..."*

DEFINITION 4.1.2: POTASSIUM NITRATE

Potassium nitrate is a chemical component denoted by KNO3, that is composed from potassium ions K+ and nitrate ions NO3-, and is a source of nitrogen. Potassium nitrate can be commonly found in rockets propellants (hence why Liz is telling her students that) and fireworks [104].

When mixing potassium nitrate with sugar, one creates gas. More particularly, that gas creates an expansion in space, which can thrust a rocket in the air!

[39] Note, since I'm not American I barely even know what a GED is, so I cannot tell you if the notion of Leidenfrost effect or the experiment that Liz is about to do is on the curriculum.

We then see rocket sketches on the blackboard behind Liz. For the purpose of my eye sight (and also because I do not know how to enhance the quality of screenshots), I'm going to assume that the rockets are, well, not rocket science and therefore everything written is correct. (On a second note, I once did a summer camp where I build a mini rocket propellant, and my rocket launch was similar to what we saw in the ep. I encourage everyone to build a rocket once in their life, it's so much fun!)

The next science bit happens at Deep Sky, where we see Kyle and Liz cure Maria's brain degeneration. I could research brain scans, and try explaining what is going on. But. We know that whatever it is, it's alien, and therefore I am prioritizing my sanity and moving on to the next subject.

Kyle tells Liz

> *"Got some new genomics software you might like to test drive. Or maybe we can see what else us interstellar internists can discover by examining Jones?"*

DEFINITION 4.1.3: GENOMICS

Genomics is an interdisciplinary field of biology focusing on the structure, function, evolution, mapping, and editing of genomes (Definition 2.4.2) [105].

A genomics software is something that indeed exists, and therefore this isn't made up science[40].

We then see Michael move-in with Alex, who says:

> *"Okay, so, now everything here is yours, except for the level three clearance laptops, um, and the sat link phone and the hard drives."*

A **satellite phone** is a phone that connects to the telephone network via satellites, instead of terrestrial cell sites (which cellphones use). Satellite phones have the particularity to be able to connect even in the most remote parts of the world (where there wouldn't be any cell service), and are also used for their privacy and security (much harder to hack into).

Alex then talks about weather balloons. Below, Alex's first line is to Michael, and the second exchange with Eduardo happens at Deep sky.

Alex: "I know that studying weather balloons doesn't sound like it's saving the world, but its research harbors some pretty cutting-edge stuff."

Eduardo: "And have you crunched the analytics for your weather balloons?"

[40] Yet. Remember that they invented a new word to basically mean genome.

Alex: "Yeah. Heat, uh, is up 1.3 degrees an hour for the last three over Roswell at large. Even though barometric pressure and wind direction dictates it should be falling. Not to mention that the ground conditions don't match up, that humidity is up 23% in the troposphere."

Now, I need to warn you beforehand, but I used to be obsessed with weather forecast as a kid, and I'm considering doing meteorology studies after I graduate. So when I first heard Alex talk about weather balloons, I got super excited, and I would like to add that studying weather balloons is super important for the world around us! Focusing on what he says to Eduardo about the measurements of the balloons, I will break down what he says bit by bit:

DEFINITION 4.1.4: BAROMETRIC PRESSURE

Barometric pressure (also known as atmospheric pressure) is the pressure within the Earth's atmosphere. The atmospheric pressure at mean sea-level is given to be 1 atm (standard atmosphere), where 1 atm = 101,325 Pa (Pascals) [106].

The higher we go, the less atmospheric pressure we found. That is explained since atmosphere is more or less the weight of air above a measurement point, and therefore the higher we go, the less air there is above us.

Note also that the barometric pressure is affected by the atmospheric density, which depends on the air's temperature and moisture (as well as the height of measurement as seen above). We then have that a low barometric pressure (when converted to mean sea level) would lead to cooler temperature and higher air humidity, whereas a higher pressure would lead to good weather.

In what Alex says here, we notice that the temperature is going up by 1.3° (I would assume Fahrenheit since it's America. But it could also perhaps be Kelvin) every hour, but that the barometer indicates that the pressure is falling, and hence the temperature should decrease. Obviously, this is not really possible, as barometric pressure is used in actual weather predictions. So clearly, Alien Science™ (which makes sense, since, you know, he's working for Deep Sky). This is also confirmed by what Eduardo says right after:

Eduardo: "Similar patterns haven't been present in Roswell since 1947."

Kyle: "Before the crash?"

Eduardo: "After."

Next, Alex mentions that humidity is up 23% in the troposphere.

DEFINITION 4.1.5: TROPOSPHERE

The **troposphere** is the lowest layer of the Earth's atmosphere, and is where most of the weather occurs [107]. We also know that humidity and temperature are related, but that the relation depends on the climate we are in.

For example, in a tropical climate, higher temperature could also go with higher humidity, which is not the case for desert climate for instance.

Once again, I don't know much about Roswell, namely the climate it has, both with regards to the show, and in real life. However, judging by the fact that Alex says that humidity goes up and he expects the temperature to go down (since he says it contradicts with the 1.3 degrees extra per hour), I am tempted to conclude that Roswell (at least show!Roswell) has a temperate continental climate.

Also note that wind carries moisture, so when Alex talks about wind direction, I assume he means that the wind is coming from a colder and more humid place, and is bringing it all to Roswell, further contradicting the raise in temperature he mentioned earlier. [41]

We have more about weather further in the episode. Dallas says to Maria:

> *"Yeah, I've been honing my connection to water lately. And I'm feeling this weird... shift in atmospheric pressure tonight."*

He says this just before it starts to rain. If we go off the assumption that Dallas was indeed talking about the purple rain (which, last I checked, was purely alien), then it simply follows to what I said earlier: lower barometric pressure gives bad weather.

Going back to the conversation between Alex, Eduardo and Kyle at Deep Sky (which is in the same scene as the conversation about the weather balloons),

Kyle: "Is that why you had me studying locusts that had shifted their migrational course?"

Alex: "You think they were affected by the weather?"

Eduardo" "Potentially."

Now, this may be purely because I am not a native English speaker, but I genuinely didn't know what a locust was before today.

DEFINITION 4.1.6: LOCUSTS

Locusts are a species of grasshoppers. The main difference between grasshoppers and locusts is that locusts can change their behaviour and their morphology (colour and shape) in response to the air density (i.e. sudden change in weather). Locusts also tend to form swarms under meteorological change [108].

The part on migration is actually rather interesting. I first found an article claiming that locusts migrate towards heavy rain and towards high grounds. Then, a decade later another article was written calling the first

[41] Please note, I am in no way an expert on the matter yet, I simply had a meteorology phase as a kid.

article bullshit and breaking down each point and how it was wrong[42] [109]. Finally, the World Meteorological Organisation said in an article that locusts migrate mostly following wind patterns, and in particular each type of locust will migrate following a certain type of wind (generally either following warm or cold winds) [110].

This then means, that if the locusts in Roswell have changed their migrating pattern, that there has been an unexpected change in wind type. However, the locusts don't migrate every time the wind changes, but only when the change is significant enough. Such a change would usually (in our world at least) indicate the coming of a storm. A big storm. Bigger than usual. Like a storm with meteorites and purple rain.[43]

4.2 FLY

[CW: 1.6K WORDS ON A SINGLE WEATHER PHENOMENON]

To save you from reading the same explanation over and over again, I will first talk about St Elmo's fire, all at once, before going to the rest of the episode. Because otherwise, every three scenes I'll have to go back to it. So, what is St Elmo's fire, and how true to reality was the show? First, here are the mentions of it in this episode, where all lines here are spoken by Liz

> *"Perfect timing. I mean, test after test confirmed it's just a case of St. Elmo's fire. Scientific St. Elmo's fire... It's an electromagnetic corona discharge that glows due to high levels of plasmic voltage that rip apart molecules in the stratosphere."*
>
> *"It's Earth science. It's not alien."*
>
> *"We saw green streaks and shimmering rain because that is how our eyes perceive ionized protons."*
>
> *"Except locusts aren't native to New Mexico. They are grasshoppers that have been genetically mutated under the right environmental conditions. So maybe this locust was affected by the St. Elmo's fire."*
>
> *"It's... Locusts, they operate on a hive mind. So if the St. Elmo's fire altered their genetic sequence, maybe it activated their swarm instinct."*
>
> *"Yeah, but it's really just... nitrogen and oxygen reacting to atmospheric electricity. Ionization... Okay, no harm in taking a tiny detour to explain the secrets of the universe. It's kind of hard to reproduce in a lab, but it operates a little bit like a Tesla coil."*

Liz starts by explaining it actually rather well, which is not surprising considering the first line we hear about St Elmo's fire is basically the first line on the Wikipedia page[44]:

[42] Call-out articles are hilarious in my opinion.
[43] Unless show!Roswell doesn't follow the same laws of meteorology as we do. Also a possibility to take into account.
[44] The sentence starts off well, and then they reformulated and now it doesn't make sense.

DEFINITION 4.2.1: ST ELMO'S FIRE

"St. Elmo's fire is a weather phenomenon in which luminous plasma is created by a corona discharge from a rod-like object such as a mast, spire, chimney, or animal horn in an atmospheric electric field" [111].

A corona discharge appears most often as a bluish glow in the air surrounding the high-voltage metal conductor. It is mostly an unwanted side effect, but has some useful applications, namely photocopying. We can find a corona discharge when attaching a metal object to a Tesla coil.

DEFINITION 4.2.2: CORONA DISCHARGE

A **discharge** (used in the electrical sense) is the release and transmission of electricity in an electric field through a non-metallic medium (most often a gas).
A **corona discharge** is an electrical discharge caused by the ionization of a fluid (here being air) that surrounds a high-voltage conductor [112].
The **ionisation** is the process of an atom or a molecule losing or gaining a positive or negative charge [113].

The best example for electrical discharge is the Geiger counter, which releases small electrical pulses that travel through air (gas) to detect radiations.

DEFINITION 4.2.3: ELECTROMAGNETIC

Electromagnetic means that we have an interaction between electrically charged particles. Usually used when talking about interactions in a field composed of both electric and magnetic fields [114].

In my opinion it isn't really useful for Liz to mention *electromagnetic* corona discharge, since the particles involved in the discharge are ionised, and therefore carry a charge. Hence, it is already implicitly said that it's an electromagnetic relation.

The next thing she mentions is "plasmic voltage". This is the part that is giving me a headache (even though at first glance it seems pretty straightforward). The thing is, that combination of words said like that *doesn't actually mean anything*. But, let's pretend, for the sake of science, that it does.

DEFINITION 4.2.4: PLASMA

Plasma is one of the four fundamental states of matter (together with solid, liquid, and gas), which contains a significant quantity of ions and / or electrons (hence, it has a significant charge). That important charge is the main distinction from plasma to the other states [115].

Plasma is the state that is the most found in the universe, and is mostly associated with stars. We can artificially create plasma by subjecting a neutral gas to a strong electromagnetic field (like inside of a corona discharge).

Since we need an electromagnetic field to generate plasma, and electromagnetic implies the presence of electricity, we indeed find the notion of voltage. We also have that plasma has a very high electrical conductivity (which physicists basically consider infinite).

To go back to the voltage, as I said earlier, we need a high-voltage conductor to generate a corona discharge (and hence to generate plasma). Since the voltage will ionise the atoms/molecules, we indeed have that plasma contains voltage but *there is no such thing as plasmic voltage*. Because the voltage present doesn't have any different properties than voltage in air, or in metal. We don't say air voltage or metallic voltage, so *why should we say plasmic voltage*.

Another thing to note is the fact that *technically* speaking, Liz is correct as to the glowing part. It's just that the vocabulary used and the way the words are put together doesn't make sense. So you don't have to scroll all the way back up, she says:

> *"It's an electromagnetic corona discharge that glows due to high levels of plasmic voltage that rip apart molecules in the stratosphere. It's earth science. It's not alien."*

Let me explain myself.

DEFINITION 4.2.5: GLOW DISCHARGE

A **glow discharge** is a plasma formed by the passage of electric current through a gas [116].

Does that definition ring a bell? A corona discharge is a glow discharge. Plasma is generated from a corona discharge, and a glow discharge is a plasma. Note that a glow discharge actually is called that because it glows. Hence, the plasma that is generated from the corona discharge is what makes it glow.

Note, St Elmo's fire is a form of plasma (for the reasons listed above). Also note that St Elmo's fire, even if it has fire in the name, is in fact not fire. We also know that St Elmo's fire is a corona discharge that happens when a strong enough imbalance in electrical charge, which causes molecules to rip apart. So again, Liz is *technically* correct.

The part that makes me italicise the word technically, is the use of "*stratosphere*".

DEFINITION 4.2.6: STRATOSPHERE

The **stratosphere** is the second layer of the atmosphere, above the troposphere (Definition 4.1.5) [117].

I haven't found anything that explicitly says that St Elmo's fire is created in the stratosphere. Since it is a corona discharge, and that doesn't necessarily require any particular layer of the atmosphere to occur, I guess this is a possibility.

But I am rather suspicious at the mention of the stratosphere here. Since the troposphere is where the weather occurs, it would make more sense to me to talk about the troposphere (not that either would be needed).

So all in all, Liz's sentence here is *technically* correct (to be taken with a grain of salt), but the wording is... not what I would use, to put it nicely. I would word it something like:

> *"St Elmo's fire is a weather phenomenon which glows due to a plasma being created from a corona discharge between molecules ripped apart."*

In the same scene, Liz 'describes' the St Elmo's fire they witnessed the night before. She describes it as '*green streaks and shimmering rain*'[45]. My theory here is that Liz is colourblind. It's the only explanation. Actually, I think it's more a case of the people doing the CGI making everything purple because that is the alien colour, and then being contradicted by the script. But Liz being colourblind is a nice Watsonian explanation.

We indeed see shimmering rain. **Shimmering** means shining with a soft, slightly wavering light. We however, do not see green strikes. We see something that looks more or less like magenta northern lights. Nothing green.

In reality, St Elmo's fire has been described as [111].

- "[...] such flashes of lightning, as appeared to illumine the whole heavens."

  – James Braid, 1871

- "Everything is in flames - the sky with lightning, the water with luminous particles, and even the very masts are pointed with a blue flame."

  – Charles Darwin, 1832

- "Butterflies became electrified and helplessly swirled in circles — their wings spouting blue halos."

  – Nikola Tesla, 1899 (when experimenting with the Tesla coil)

- "The whirling giant propellers had somehow become great luminous disc of blue flame."

  – William L. Laurence, 1945

- "The water, like a witch oil's, burnt green and blue and white."

  – Samuel Taylor Coleridge, in *The Rime of the Ancient Mariner*, 1798

These quotes have all be said by people who actually witnessed a St Elmo's fire. As you can see, it's mostly under the form of a blue lightning strike, or like a flame dancing on a conducting object. Not like purple

[45] The full quote is given above, but I am only focusing on the visual elements here.

northern lights. And not quite like green streaks either. So really, Liz is so far off, it's actually quite impressive.

The science behind St Elmo's fire's colour can actually be explained. It is due to the nitrogen and the oxygen in Earth's atmosphere, which make St Elmo's fire fluoresce with blue or violet light. We can also compare the colour to a neon light, as it is the same mechanism (the colour is different because of the different gases).

So, on this front, Liz is actually rather correct, when she describes it as "green strikes", even if blue would be a more appropriate colour. And the CGI isn't far off with the colour either, since St Elmo's fire does glow purple (just not like a northern light, and in a bluer shade than the magenta they showed).

Also, side note, but I understand why no one believed Max when he said it was a real-life science phenomenon. He described something green, to people who saw something purple. Not quite the same.

The next thing we have about St Elmo's fire is when Liz mentions the locusts, where she mentions that the St Elmo's fire could have activated their swarm instict.

This also correct, locusts (Definition 4.1.6) are a mutated version of grasshoppers, and their mutation is due to the density of the environment they are in, which in itself is due to the weather.

Then, when talking to her students, Liz explains how St Elmo's fire works: '*nitrogen and oxygen reacting to atmospheric electricity, ionisation...*'.

That is indeed correct. See above for all of the explanations. She then tries to demonstrate how it works, except she doesn't quite reach a corona discharge. I will say though, had she not interrupted herself when Vanessa leaves, she *should* have gotten the correct result. Liz says: "*It operates a little bit like a Tesla coil.*"

DEFINITION 4.2.7: TESLA COIL

A **Tesla coil** is an electrical resonant transformer circuit designed by Nikola Tesla. It is used to produce high-voltage, low-current, high frequency alternating-current electricity [118].

It is more or less what we see Liz do. And as said above, we can produce a corona discharge by attaching a conducting metal rod to a Tesla coil, which is what I suppose what Liz was going to do.

And that's it, this is all I have to say about St Elmo's fire. I am rather quite sad that they didn't really talk about it afterwards, but I guess it wasn't really an important plot point.

The next chronological science fact we have is Liz sequencing (Definition 2.4.1) the locust's genome (Definition 2.4.2) in only a few hours. Very much not true. It would actually take a week or more. Then, Liz and Michael dissect the locust, and talk more about the species genetic mutation (the same dialogue as earlier, but I am now going to focus on the locusts).

Liz: *"Except locusts aren't native to New Mexico. They are grasshoppers that have been genetically mutated under the right environmental conditions. So maybe this locust was affected by the st Elmo's fire."*

Liz: *"Locusts they operate on a hive mind. So if the st Elmo's fire altered their genetic sequence, maybe it activated their swarm instinct."*

I know nothing about what species are native or not to New Mexico, and I didn't really look into it for 4x01, so I thought I'd look into locusts and New Mexico for 4x02. And guess what?

There is a species called the Carolina Locust (*Dissosteira carolina*), which is native to the lower 48 United States of America, except for southern Florida, Gulf Coastal Plain, bottom 2/3 of California, and southwest Arizona. For those of you like myself who don't know, the lower 48 states refers to the states excluding Alaska and Hawaii. I know I'm not a geography expert, but I had to learn all the US states in school. And I'm pretty sure that New Mexico is not in Alaska or Hawaii.

Another fascinating fact about the Carolina Locust, is that if you go on Google image, it looks suspiciously like the locust that Liz and Michael are dissecting. The one who isn't native to New Mexico, apparently. This is actually the only thing that is factually incorrect, the rest is either true, or alien science.

Finally the last science fact of this already scientifically heavy episode is once again when Liz is teaching to her class. It is also the scene where I cannot unsee the typo.

I have already talked about Liz experimenting with the Tesla coil, so I will now focus on the board behind her. On the board we see that the students are studying noble gases. A Screenshot of the board is given in Figure 4.2.

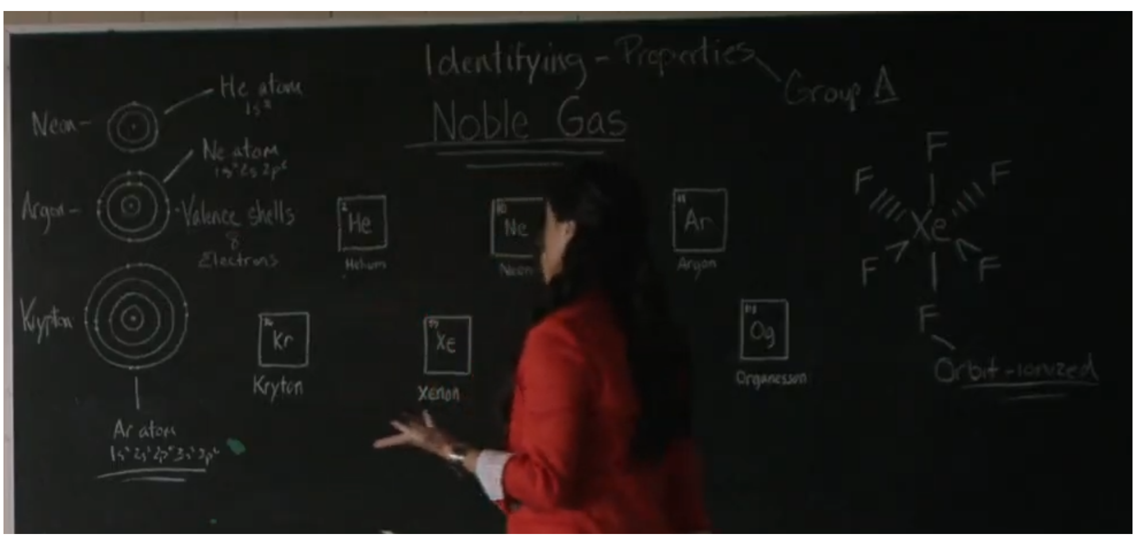

Figure 4.2. Screenshot of the board with Liz, taken at 20:11.

What is written on the board seems to be correct, except that there is a typo. The element Kryton doesn't exist, it should be written Krypton instead, which is indeed a noble gas. Since the quality is rather low, I have added a sketch of what I could decipher from the board in Figure 4.3, where I have also added corrections to the typos that the appear. Namely, they have Organesson as an element, which does not exist and should be Oganesson, and they have the wrong electron configuration for Neon, which should be [He]$2s^2 2p^6$, or when substituting the electron configuration for Helium, $1s^2 2s^2 2p^6$. I also corrected the Valence shell for Argon (the third one). There is one element on the board that I was not able to identify since Liz is always standing right in front of it, but given that the board lists all noble gases, there is only one element missing, namely: Radon (Rn) with atomic number 86.

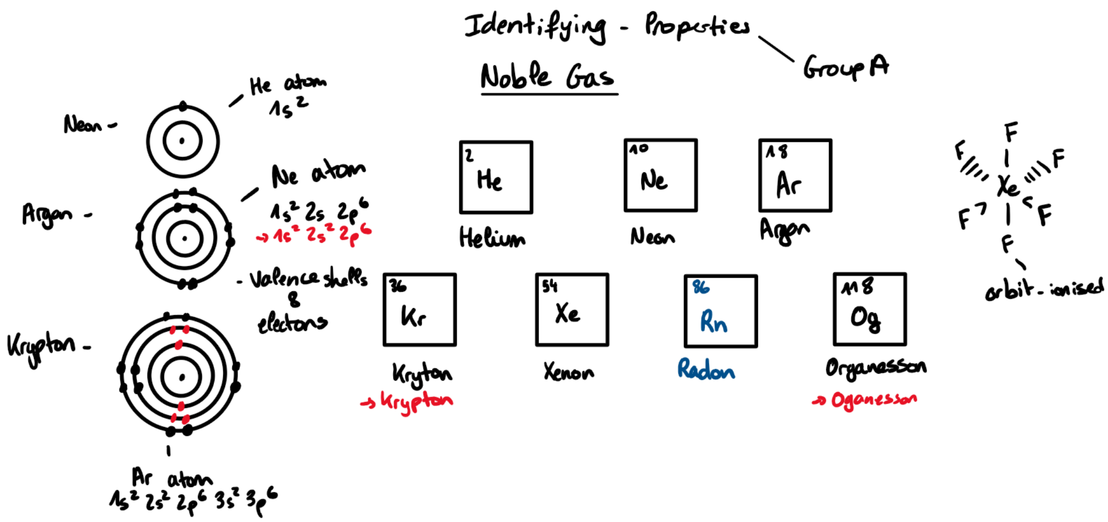

Figure 4.3. Sketch of the board from Figure 4.2 (with corrections in red, and my guess as to element that cannot be seen in blue).

As a side note, I am a little bit confused by the information on the left side, where the Valence shells drawn correspond (respectively top to bottom) to Helium, Neon, Argon. However, I do not really see why Neon, Argon, Krypton are written on the left of the shells. The drawing on the right corresponds to a Fischer projection [119] of a Xenon hexafluoride [120] (F is Fluor, with atomic number 9).

DEFINITION 4.2.8: NOBLE GASES

The **noble gases** is a class of chemical elements that all have similar properties under standard conditions: colourless, odourless, monoatomic (i.e. contains a single atom) gas, with a low chemical reactivity.
There are six of them: helium, neon, argon, krypton, xenon, radon (the only radioactive one), and sometimes this list also includes oganesson [121].

Liz makes a joke about "tea with xenon?" This is because xenon is used as a general anesthetic. We also see on the board "Valence shells".

DEFINITION 4.2.9: VALANCE ELECTRON

A **valance electron** is an electron on the outer shell that helps to form a liaison between atoms and molecules [122].

This episode also has a lot of alien related science, which for the sake of my sanity, I decided not to dive into.

4.3 SUBTERRANEAN HOMESICK ALIEN

Compared to the last two episodes, this one is very light on science. The two science-y bits happen at Deep Sky (can we talk about how everyone seems to just walk in and out of that place? including dead bodies). The first one is when Liz thanks Eduardo for giving her access to his machine.

> *"Eduardo, thank you. This machine is my favorite thing. It tests for everything: EC, pH, PO4-P. I'm about to ask it to do my taxes."*

I will break down all these initials one at a time.

EC could mean a multitude of things, even when just looking at biology and chemistry [123]. I will spare you all the different possible definitions and directly give you the two options that I think Liz might be referencing:

- **EC50**: which is the half maximal effective concentration, and is a toxic unit measuring the concentration of a drug, antibody or toxicant. To put it in simpler terms, EC50 is the concentration required to reach 50% effect [124].
- Or she could be talking about the electrical conductivity of a solution. Which, as the name indicates, represents the ability of a certain material to conduct electric current [125]. The other options would not make that much sense in the context of the sentence.

I would be more inclined to believe that she is talking about EC50, especially given she talks about pH just after.

DEFINITION 4.3.1: PH

pH (potential of hydrogen) is a measure of how acidic/basic a solution is. The range goes from 0 - 14, with 7 being neutral. If the $pH < 7$ then the solution is acidic, and reversely if the $pH>7$ then it is basic [126].

Fun fact, but testing for pH (as far as my chemistry knowledge from high school goes) *is not that hard*. And I don't think it requires a special machine to do. The way we did it in high school (granted, we were not doing very complex solutions), is that we stuck a piece of pH paper into the solution, and depending on the color we had the scale of acidity.

The third thing Liz mentions is PO4-P. **PO4-P** is a method of testing for phosphorus. It basically gives back the number of phosphorus (and only phosphorus, not oxygen) present in a solution [127].

We know from 2x01 where Michael and Liz are dissecting Noah's body, that Oasians' organs (other than bones) contain traces of organic phosphorus (Definition 2.1.1).

I tried finding a link between phosphorus and P2P (phenyl-2-propanone) (Definition 1.1.1), and the closest I got was that both P2P and phosphorus can be used in creating meth (but they are both different methods, and it requires mostly red phosphorus) [128]. So, the logical conclusion would be that the mystery component is both a derivative of P2P and of phosphorus.

Either way, we do know that alien organs have stronger traces of phosphorus than humans (hence why the little green men typically glow in the dark), and since Liz's machine can test for the presence of phosphorus, we now have an easier way to track down aliens and their environment.

The next bit of science follows in the next scene, where Liz tells Max (after having analysed all the samples in record time. Really, Deep Sky is like heaven with how fast things happen):

> *"All right, well, the alien plasma matches the balloons, but that's to be expected since the samples we took were from their car wheels. Usual levels of bacteria and fungi, EC is normal. Huh... High levels of petroleum hydrocarbon."*

Once again I think EC is actually EC50 as I talked about earlier, since I don't think electrical current is relevant here. And now, petroleum hydrocarbon. Max says that it's the result of oil extraction. That is correct.

DEFINITION 4.3.2: PETROLEUM HYDROCARBON

Petroleum is a naturally occurring yellowish-black liquid mixture of mainly hydrocarbons [129].
A **hydrocarbon** is an organic compound consisting only of hydrogen and carbon [130].

The thing that bothers me in that scene, which I've mentioned before, is the speed at which the science is done. I understand that actual science consists mostly of waiting and therefore isn't interesting to show on TV. However, if everything was simply a little more spread out, it would make a lot more sense scientifically speaking.

4.4 DEAR MAMA

We start the science in the episode with a dialogue between Isobel and Liz.

Isobel: "All right, uh, Dr. Dyson, why don't you tell me about this, uh, alien ectoplasm, hmm?"

Liz: "Oh, I prefer the term 'goo', but it's technically epidermal tissue."

DEFINITION 4.4.1: ECTOPLASM

The **ectoplasm** is the outer layer of a cell's *cytoplasm*. The ectoplasm is usually of an elastic texture [131].
The **cytoplasm** (Definition 1.8.4) is the all of the material in an *eukaryotic cell*, enclosed by the cell membrane, excluding the *cell nucleus*.
Eukaryotic cells (Definition 3.9.1) are cells whose *nucleus* is enclosed by a nuclear envelope.
The **cell nucleus** is what contains all of the cell's genome (all genetic information of an organism) (Definition 2.4.2) [132].

So, the thing is, what we see on screen looks like skin that has been shed. Which is not what ectoplasm is. And I really don't know why Isobel would mention that. I guess one could theorise that after spending many hours around people (Liz and Kyle) who talk about biology, one would absorb some of that vocabulary.

However, Liz indeed corrects her, by saying something scientifically correct! Skin is indeed epidermal tissue.

DEFINITION 4.4.2: EPIDERMAL TISSUE

The **epidermal tissue** is the outmost cell layer of the primary plant body [133].

So, really, humans are plants and therefore the outer layer of skin, since our skin is composed of 3 layers, one of which is called the epidermis (corresponding to the outer layer), which is an other word for epidermal tissue.

I know this is a weird thing to be proud of, but the fact that they used correct terms (as far as I know) is pretty cool. Just afterwards, Liz mentions the third helix and hence it is definitely alien. Which, personally, I didn't need to look at it under a microscope to tell. The skin sample glowed. Human skin, as far as I know, doesn't do that.

But, better be safe than sorry. It's always good to check your results, and give background to the results that were obtained.

The next science is something that confused me at first, and then seemed to make some sense after some amount of research. Liz seems to be walking around the desert with a metal detector (spectrometer). And she says

> *"Yeah. I rigged it so that it would locate the alien tech component in alien biology. I mean, at least I hope I did."*

DEFINITION 4.4.3: SPECTROMETER

A **spectrometer** is a scientific instrument used to separate and measure spectral (over a spectrum) components of a physical phenomenon [134].

There are light spectrometers, that can separate white light (visible light) and measure the individual bands of colour that compose the light (called a spectrum). There are mass spectrometers which measure the different masses of the atoms or molecules in a gas.

There are other types of spectrometers, like a magnetic one, which is what I had initially assumes Liz had used, but turns out a magnetic spectrometer requires very specific conditions.

Then, I decided to look further into mass spectrometer. It appears that it would work on solids rather than just on gasses. We then can conclude that Liz probably used a mass spectrometer, and tweaked it so that it would "detect" the change in mass between alien DNA (which contains a third helix, which I'm guessing must affect it's mass), from the rest of the desert.

Well, she does find an odd change in mass, in, you know, a truck buried under sand. Very normal, of course[46].

We then have more Liz and Kyle science!

Liz: *"Kyle. What if the alien cells aren't dead? What if they are only playing dead? Like an octopus does when it-it's scared and it turns into coral."*

Kyle: *"You're talking chromatophores."*

Octopuses[47] can indeed turn into coral! It's fascinating to see[48]. Squids can also change their colour[49]. Squids and octopuses have cells called chromatophores that have the particularity of changing colour. These cells are situated just below the the surface of the skin, which is thin enough to show the colour of these cells.

Chromatophores work by having a center that is elastic and contains a certain colour. The elasticity means that when it stretches out, the colour would appear brighter, and hence give the illusion of changing colour (since then the colours seems more visible). This elastic center, like a little pouch, is controlled by nerves and muscles. These cells are found in a large number, in varying sizes. Hence, it can create a full camouflage. Thing of it like pointillism, a painting made of thousands of little dots of different colour [135].

To go back to the alien science, basically the alien cells taken from the skin sample have the same property than squids and octopuses. By introducing the sample from the blue eyes, it is like saying "hey don't be afraid, I'm a friend", and the cells then revert back to their "original form".

[46] Liz saying "For the first time in forever, the science doesn't make any sense to me." really really describes my constant state watching that show

[47] I had to google the plural of octopus, and I'm just going with what google answered

[48] Here is a YouTube video for you.

[49] I can talk about squids a lot, I have done extensive research on them

4.6 KISS FROM A ROSE

Michael is sick and his powers are gone. The first theory that Liz emits is:

> *"The regenerative protein that keeps you from getting sick is the same place where you derive your powers."*

Meaning that when Bonnie kissed Michael, she would have "deactivated" that protein, both inhibiting his powers, and rendering his immune system useless. Hence, he would then be affected by every bacteria surrounding him.

Then, at Shivani's lab, Liz finds out that in fact, Michael seems to have an alien virus. Shivani is the one to point it out, that the virus has a triple helix. This then means that Bonnie would have taken Michael's powers away by giving him some sort of alien virus which would then enter the regenerative protein that gives aliens their powers.

The next science scene is once again between Liz and Shivani. They reach the following conclusion about Michael's illness:

Liz: *"This virus was clearly built to replicate inside of a host, but it's not replicating in this sample."*

Shivani: *"Which means your friend can't spread it."*

I personally find it quite worrying that up until that point, they didn't know if Michael could spread the virus, and still didn't take any protection. It is a know fact that a virus can mutate, and attack people who don't even seem at risk (just look at the covid, it's a perfect example). You'd think that everyone would take precautions not to spread whatever Michael has, while still making sure he is not alone.

On a second note, the part of "this virus was built to replicate inside a host but it's not in this sample" is very interesting to me. (I will get back to it when I talk about Clyde stealing powers more in details. This will either be in the science meta dedicated to that episode, or in the alien powers meta for that season; I haven't decided yet).

Liz basically says here that the virus lives inside of Bonnie; that it replicates regularly, and by extension (I think) it is safe to say that it then produced by the protein that regulates alien powers. It is then a personal ability; the virus is only replicating in Bonnie, in someone else it will only attack the host, but not try to expand into other people (unlike covid and most of other viruses).

There are viruses that we know do not spread easily, but there is still always a risk of contagion. Which circles back to my earlier point about Michael not being a some sort of quarantine.

When Michael confronts Bonnie at the police station, she says "*You're sick? That's never happened before.*" That is a very interesting sentence. Because we know that Bonnie has infected other people, taken away their powers (her mother has she later tells us). But the fact that Michael is the first to get sick is due to the fact that they are on Earth and not on Oasis. (This is not confirmed but I really don't see any other option.)

We can then conclude, or at least hypothesise, that Oasis is not an environment in which bacteria and viruses would survive easily (if at all). Because if this was only due to the alien DNA, it would follow that the people whose powers Bonnie took would fall sick too. Which doesn't seem to be the case.

The fact that there are less viruses on Oasis could be either a technological, a biological, or an environmental improvement. We know that alien tech and alien biology have close similarities, so it is not that far a stretch that the environment on Oasis would be somewhat similar.[50]

We then see Liz stab Tezca with the machete.

> *"Which is why I tipped that machete in Bonnie's virus. Then I added a chemical accelerant to speed things up a little bit."*

We aren't given a lot, but I will assume that Liz used the samples she had from Michael to separate the virus from his cells. Which is possible, but the different techniques depend on whether the virus has a single-stranded or double-stranded RNA. Here we have a triple helix virus, which we do not have (yet) in our world[51].

Then, another Shivani science scene. She hands a woman a vial and says

> *"The regenerative protein we've been looking for. It's damaged, but it's a start."*

I am guessing she got that protein from Liz, who took it from Michael[52].

I do not know whether it is possible to repair proteins from a sample that have been damaged by a virus. I have found a few articles on the regeneration of proteins after a virus in the human body, but as for outside of a body, i.e. in an artificial, lab-like environment, I haven't found much that I could actually understand.

For the sake of the show, I will say that Shivani has hired a ground-breaking scientist who does in fact know how to repair proteins artificially.

4.7 DIG ME OUT

During diner at the restaurant with Liz, Rosa and Shivani, we have the following dialogue between Liz and Shivani:

[50] Side note but Bonnie saying "I can't use the powers I take" and then cut to Clyde using said powers is my villain origin story.
[51] I am not a biologist, even less a virologist, so I won't even try to understand the science behind this. I just know it's a thing.
[52] For the sake of my sanity, I will ignore all the ethical problems in this episode – in this entire show even.

Liz: "Wait, so... we were looking for a way to attack damaged tissue. Right? [...] Okay, but fractals, they're all over the place. In cell physiology, and in human anatomy. I mean, what if we used that same pattern?"

Shivani: "We might be better able to holistically mitigate damage."

Liz: "Not only that. We could regenerate the cells altogether. I mean, you have this new cryogenics company, right? An immersive chamber might be exactly what we need to undo the damage that has altered these cells. We could flood our subjects' circulatory systems with bio-engineered organic compounds that are in a cryogenic..."

Shivani: "Pod"

So. Let's begin from the start. As Liz says, we indeed have fractals everywhere. And I would be a bad mathematician if I didn't start by talking about fractals in maths (this is purely an excuse to nerd out over maths and forget about biology for a moment).

DEFINITION 4.7.1: FRACTAL

A **fractal** is a geometric shape containing detailed structure at arbitrarily small scales. Many fractals appear to have a similar pattern at increasingly smaller scales[a].

[a] Like the Mandelbrot set, for which I made a gif.

Fascinating isn't it?! This is just one of the most famous examples of fractal patters. Fractals don't just appear in mathematics, but also everywhere around us! For instance, some species of algae, lightning bolts, and DNA all contain fractals. So this was the fun part of that conversation with Liz and Shivani. Now, let's have a look at the less fun part: "We might be able to holistically mitigate damage."

DEFINITION 4.7.2: HOLISTICALLY

Holistically (in medicine) means "in a way that treats the whole person, taking into account mental and social factors, rather just the symptoms of a disease." [136]

So, here, Shivani is suggesting that they might be able to (using fractals) reduce the damage on the whole person (i.e. not just their body). Which, when it comes to minimising damage in medicine is not something that is always so easy to do. So, fractals would (according to Shivani) allow them to minimise the damage done to Shivani's daughter and Max on a global scale.

Then, Liz mentions that they could use fractals to regenerate cells.

> *"We could regenerate the cells altogether. [...] We could flood our subjects' circulatory systems with bio-engineered organic compounds that are in a cryogenic..."*

Is regenerating cells using fractals something that is scientifically possible? According to some papers that

I've read, the answer it yes. The idea behind this method is that since cells have a fractal pattern to them, we can use fractal dimensions, and their repeating pattern at the wanted scale, to influence how new cells grow. This can be done by introducing vasculature (= vascular system of a part of the body) to the target cells. The scientists use computer programs to recreate that vasculature, by generating the wanted fractal pattern to insert to the cell.

So, this was a little explanation of how fractals can be used in the regeneration of cells. Which, is definitely something that Shivani would have the resources to do and to finance. Now, let's look into the next science fact mentioned in this conversation:

DEFINITION 4.7.3: CRYOGENICS

Cryogenics (in physics) is the science behind the production and behaviour of materials at a very low temperature.

The special branch of cryogenics applied to biology is called cryobiology, which usually has the purpose of cryopreservation. As far as I can read online, the use of cryogenics in cell regeneration is mostly about preserving the cells in their current state.

So, in terms of Liz trying to regenerate Max and Michael's alien cells, my guess is that she could freeze their current cells so that they don't degrade more, before finding a cure? To be honest, all of the papers I've read about cryogenics and cells talk about preservation, so I can't really see any other use.

However, for Shivani's case, cryopreservation is very accurate, for the lack of a better word. Her daughter is in a cryogenic pod (as seen in 4x06, and as mentioned by Shivani in the above conversation), which mimics the preservation of the alien pod, by keeping her cells in stasis.

So, in conclusion, in my opinion, the use of cryogenics makes sense when we consider Shivani, but not when we consider Liz. And since Liz does not know about Shivani's daughter, I will assume that this is Liz's way of keeping the alien secret, by not revealing that she too possesses a pod, only not a cryogenic one.

4.8 MISSING MY BABY

The first mention of science is Liz talking to Maria in the Wild Pony: "*There has to be a scientific explanation for how Alex is reaching out to you.*"

She then proceeds to explain her *scientific theory*.

> *"At first, I thought this was a resurgence of your powers, like a vision of Alex manifesting itself. But your brain scans match up with your last checkup, post-cure, which means this is not a vision. Alex is alive. And he's figuring out a way to communicate with you. We've seen versions of this before. You did it in the coma. Max talked to Rosa in the pod."*

I should probably go ahead and say that this falls in the realm of science-fiction. Well, mostly. In reality, people in a coma can 'communicate' with the outside world. More or less. Studies have shown that when talking to people in a vegetative state, asking them to perform certain tasks, the brain reacts in the same way it would for someone that is not in vegetative state. For example, when doctors asked a few yes-no questions to a patient in vegetative state, they found out that he knew where he was, what kind of program he enjoyed on TV, etc [137]. This was done by monitoring the brain activity of the patient and looking at the pattern in his brain activity. This kind of research allowed scientists to conclude that some comatose patients were conscious [138].

Going back to our lovely little alien show, we can argue that Maria communicating with Liz while in her coma is scientifically possible. However, Alex being in another dimension and communicating through with Maria is fiction. As a side note, Liz mentions a few time in that scene that it's all science, except it's not.

Next, Liz and Shivani are doing science in Shivani's lab.

Liz: "The regenerative enzymes in the blood sample we brought back from New York isn't healing the diseased cells in our subject."

Shivani: "What about using the bio synthetic compound that my team developed?"

Liz: "The fluid stimulates the proteins momentarily, but it isn't potent enough to sustain their viability."

As defined earlier (Definition 1.8.5), an enzyme is a protein that accelerates a chemical reaction. Here, regenerative enzymes refers to enzymes that have the ability to regenerate their cofactors.

DEFINITION 4.8.1: COFACTOR

A **cofactor** (in biology) is a non-proteinic chemical compound that is required for enzymes to act as catalyst (i.e. to accelerate a chemical reaction) [139].

Now, from what I read on the internet, enzymes are used in cellular apoptosis, which is a programmed cell death, happening naturally in our body (Definition 2.4.4), and the regenerative aspect of enzymes comes into play with regards to their cofactors. Furthermore, I have found a few sources about regenerating dead cells (namely after injuries) that mention enzymes as 'executioner' rather than reconstructive. In other words, it appears that enzymes (regenerative or not) are present when it comes to killing cells, but not resuscitating them [140, 141].

So, really, it's not that surprising that Liz's enzymes aren't healing the deceased cells...

The next bit of science happening in Shivani's lab is purely fictive as it revolves around the cracked pod that Shivani has:

Shivani: "The viscous liquid we developed was formulated using the fluid from within [the pod]. It contains restorative properties identical to the ones that you've been researching."

Liz: "In gene-sequencing your work, I found micromolecular inconsistencies, and this crack explains why. Oxygen is changing the fluid's chemical composition. In order to continue testing, we need a pod that's in working condition."

I may be wrong here[53], but I believe that so far we have been told that the pods have a conservation property, not restorative. Particularly since Isobel went into the pod not as a way to evacuate the serum she injected herself with, but as a way to give Michael and Liz time to come up with an antidote. However, one could argue that in order to conserve the cells of the person inside the pod, it would have to regenerate the cells that are dying (or be able to pause the evolution of the cells, either way).

Personally, I am not a big fan of the pods being the answer the everything, namely here providing a regenerative cure, especially when we saw Liz figure it all out using science before, and since her research is about vascular *regeneration*.

The part on oxygen altering the fluid's chemical composition is accurate to what we saw before, with Noah's pod. Hence, this would fall into the realm of science within the fiction, since at least the results are consistent.

We then see Rosa and Maria trying to communicate with Alex at the Wild Pony. This is once again clearly fiction, but they use science words that I thus will look into. We start with (another) mention of frequencies:

Rosa: "I can hear the frequency of Alex's DNA in the hair, but... nowhere else."

At first, the mention of frequencies in the show was very nice and (mostly) accurate. Then, it became too much.

DNA does have frequency, and it particular it generates a wave that propagates in the direction of the magnetic field vector [142]. **Magnetic field vector** refers to the direction and the dimension of the magnetic field. These magnetic waves are what allow the cells to communicate with each other.

It is also possible to detect the frequency of DNA, namely it is around 20 GHz. Note that 20 GHz falls into the domain of super high electromagnetic frequency, which is the same frequency range as satellite communication and microwave ovens. Suppose Rosa can hear the frequency of Alex's DNA in his hair. This would mean that she is able to hear satellite communication happening nearby, and considering there is a military base not too far away (I don't know anything about actual Roswell geography, but the show does mention the base a few times), she would be able to pick up on those conversations. Useful espionage tool.

Rosa does mention something about Isobel teaching her how to listen: "*Isobel taught me that in order to silence all the noise, you have to find a place inside yourself where you feel safe and calm.*"

[53] It's been a long time since I watched the first season 1.

And in real life, radios can tune to a particular frequency (hence how you only hear a single radio channel and not everything at once). However, a radio is also able to detect other frequencies, hence how you can switch channel. The reason why you do not hear the other radio channels when you switch on your radio is that radio emitters in a region have an agreement to distinguish each frequency. This means that the frequency of each channel has to be sufficiently distinct.

DNA is different for everyone, hence it's frequency differs. Since Alex, Maria, Rosa, and any one else who came to the Wild Pony are not twins, Rosa would be able to distinguish Alex from the other DNA frequencies.

However, there are three things I find strange. First, I am assuming Alex was at the Pony before disappearing, so there should be traces of his DNA left, hence Rosa would hear it *somewhere else*. Secondly, the choice of detecting DNA at the Wild Pony is weird, in my opinion, since there are many more frequencies that Rosa has to tune out, as it is presented to be a popular bar.

Thirdly, as much as DNA can be extracted from hair, there is no guarantee that DNA is *always* available in a hair sample. Namely, the DNA is present at the root of the hair [143]. Without the root of the hair, the procedure to get DNA is much more complex. So, for the sake of the argument, I am assuming that Rosa has the root of Alex's hair.

But, even having the root of the hair, typical DNA testing that is done in forensics requires an electron microscope (which uses a beam of electron as a light source, and thus has a better resolution than a typical light microscope [144]), since it is the *only* way to visualise the small bits of DNA. The second test is through mitochondrial DNA, but such a test will not distinguish between siblings, as they share the same mitochondrial DNA profile. DNA testing from hair also requires more than 1 hair to have an accurate test.

To conclude on the topic of hair DNA and frequency, despite DNA having a frequency, and hair having DNA, I doubt the accuracy of Rosa's test, since too little DNA is present in a single strand of hair.

The next bit of science is also said by Rosa:

Rosa: "Energy rises from the ground into your feet, and since consciousness is a tangible form of energy connecting us to all matter, astrophysicists believe that... meditation could open you up to speaking over universes, even dimensions."

I am not going to touch on this topic, other than say that Rosa probably meant astrologists and not astrophysicists. Astrology is the field of divinatory practices, recognised as a pseudo-science (it used to be a science until a more scientific theory came along), while astrophysics is the study of astronomical objects using physics and chemistry. Meditation falls into the category of astrology, not astronomy.

The next two pieces of science (the last ones for this episode) are all between Liz and Shivani, during different scenes.

Shivani: "Why atomize the fluid into a mist?"

Liz: "I made slight genetic modifications to accelerate enzyme inhibitors for swift cell regeneration. Molecules, they travel faster in a gaseous state than in liquid form."

Shivani: "If this cure eradicates the alien virus, could it be repurposed to fight off human illness?"

Liz: "Like a universal vaccine? Interspecies scaling at this level would be a huge leap."

So. Liz's first sentence is already hurting my brain.

DEFINITION 4.8.2: ENZYME INHIBITOR

An **enzyme inhibitor** is a molecule that attaches itself to an enzyme and stops its activity [145].

Since enzymes are used in process of cell death, inhibiting its activity would then slow this process. In fact, inhibitors can be used to start the process of cell regeneration, in some circumstances, although I am not sure I understood what those circumstances were precisely [146]. So, if Liz accelerates the inhibitors, it could lead to an acceleration of the regeneration process (or at least, make it begin faster).

It is true that molecules travel faster as gas than liquid. In fact, molecules are always moving, but in a solid they are moving so slow that it can be considered static. In a liquid, the molecules glide past each other, and hence are faster in movement than in a sold. In a gas the molecules are further apart from each other, and move faster [147].

Shivani then asks if a vaccine to cure all human illnesses is possible. Liz is right with her response, that interspecies vaccines are tricky to work with.

We first need to note that interspecies contamination is possible (coronaviruses are example of this). However, interspecies vaccination is not as simple; consider the infectious bronchitis virus (IBV) in poultry, for which different vaccines have to be created for the different species of poultry. Even species which are similar (poultry for instance) would reach very differently to the same vaccine, and would not be offered the same protection [148]. So, despite humans and oasians being similar, a vaccine that would protect both is also unlikely.

Furthermore, a universal vaccine in the sense of a single vaccine that would protect against everything in scientifically improbably. The point of a vaccine is to inject into the human body a small dose of a virus to teach the body how to defend itself against it. Therefore, a universal vaccine would have to consist of a small dose of every virus known to mankind, which would very likely kill the patient. Imagine having your body fight against all the viruses ever.

So what Shivani is suggesting is rather improbable.

The final bit of science is later in the episode, still in Shivani's lab.

> *"What if we altered the protein receptors to perform some, not all, of their functions? If we folded the principle alpha helices and stabilized the covalent bonds between the amino acids..."*

If I am being honest, the sentence "altered the protein receptors to perform some, not all, of their functions" confuses me. So I am ignoring it for the sake of my mental health (also because I don't understand what the 'their' is referring to).

The next bit of science that Shivani mentions is "principle alpha helices".

DEFINITION 4.8.3: ALPHA HELIX

An **alpha helix** (α-helix) is a sequence of amino acids in a protein that are twisted in a helix shape. This structure is the most common type of secondary structure in proteins [149].

In everything that I've read online about alpha helices, there isn't really a notion of *principle* alpha helix. So, I don't really know why this is specified by Shivani here (probably to make the sentence sound science-y...).

DEFINITION 4.8.4: FOLDING

Folding in chemistry is the process by which a molecule assumes its shape. Protein folding refers to the shape that is assumed by a specific sequence of amino acids in a protein [150].

So, from what I understand, folding the protein would give it its alpha helix structure, but folding the alpha helix structure doesn't really mean anything...

DEFINITION 4.8.5: COVALENT BOND

A **covalent bond** is a chemical bond that involved sharing electrons to form electron pairs between atoms [151].

The different structures in a protein are attached via covalent bonds [152], so indeed, there are covalent bonds between some amino acids (the ones that are found in two different structures). Alpha helices are considered to be the stabler structure of a protein, but their stability, from what I've read, is not quite due to the covalent bonds. So, again, despite the different elements of the sentence making sense, they are quite confusing and lack proper meaning when being put together

4.9 WILD WILD WEST

The first science word in this episode is said by Michael, talking about how to help Liz:

> *"We need to find the naloxone."*

DEFINITION 4.9.1: NOLAXONE

Naloxone is a medication used to reverse or reduce the effects of opioids, by reversing the decline of the nervous and respiratory systems due to opioids [153]

As Michael says, according to Kyle, it will reverse the effects of the drug Liz took. However, as Rosa points out, naloxone will reverse the effect of opioids, there is no guarantee that it will work on Liz, as there is no guarantee that the alien mist that Liz inhaled resembles opioids.

Further in the episode, Heath shows up and starts talking about science. He explains all he does, and there really isn't much for me to add since he says it all himself:

> *"Well, I can't be sure until I run some more tests, but... cold hands, fever, rash– it could be autoimmune. [...] Meaning that the mist could be causing her body to attack itself. I'm gonna give her some steroids. It should suppress the immune response. Well, at least, in normal circumstances."*

After injecting the steroids, nothing much happens, so Heath goes to the lab to cook something stronger. As Michael said, "alien toxins call for alien cures". What I love about Heath is that he explains what he does, and he doesn't say made up words. The next thing he says is:

> *"Well, the drug has latched on to Liz's nerve receptors. Somehow it's managed to trick her immune system into thinking that her own nerves are actually foreign bodies. Which would be an easy fix, if I had a way to camouflage alien proteins inside human biology. Unfortunately, NYU hasn't quite gotten there yet."*

He establishes the problem, explains his solution and he mentions how human science cannot solve alien problems. What a great guy.

4.10 DOWN IN A HOLE

[tw: mentions of animal testing, dead animals, blood]

The first bit of science of this episode is, to my despair[54], more talk about frequencies. This conversation is between Rosa and Maria.

Rosa: *"Well, Heath said that what you're sensing might be similar to my abilities. Everything has a resonant frequency, right? And I'm guessing so does the liminal space that our friends are stuck in."*

Maria: *"Wait, so you're saying that my new powers might be picking up frequencies in other places, too?"*

Rosa: *"Exactly. And as you tap into the energy rippling off that you're able to get glimpses of our friends who went all Land of the Lost. It's something that the CIA tried to do, is bend time and space to spy*

[54] Although I do enjoy it, as you can probably tell by the long text below

on their enemies. They called it remote viewing. But they didn't have a real psychic."

I have already talked quite a bit about frequencies with regards to this show and the alien powers (Section 3.7 the conversation between Michael and Rosa about her powers, quite a collection of accurate scientific facts, and Section 4.8 where Rosa talks about hearing the frequency of Alex's hair DNA, still technically scientifically sound, but a little far-fetched), so I will not delve too deep into it again[55].

Yes, everything does have a frequency. Yes, if you can pick up on a certain range of frequencies, then you can pick up waves that come from somewhere else (think of it as FM radio, you can pick up a few different frequencies from a few different places). However, typically, a receptor picks up a range of frequencies, and waves that are not in that range are not picked up (or not on purpose that is).

Now, Rosa also says that if Maria "*taps into the energy rippling off [of the frequency], then [she'll] get glimpes or [their] friends*". Breaking this down, frequency and energy are indeed related. Namely, Planck's relation states that the photon energy E is proportional to the photon frequency f, such that $E = hf$, where h is the Planck constant [154].

DEFINITION 4.10.1: PHOTON ENERGY

A **photon** is a massless elementary particle that moves at the speed of light. Photons are present in light or radio waves, at different energy levels [155].

The **photon energy** is the energy carried by a single photon, typically measured in electronvolts (eV) [156].

DEFINITION 4.10.2: PLANCK CONSTANT

The **Planck constant** h is a fundamental physical constant, with a value of (about) $h = 6.6x10^{-34}\text{JHz}^{-1}$ (Joule per Hertz) or $h = 4.1x10^{-15}\text{eVHz}^{-1}$ (electronvolt per Hertz). This constant determines (together with some other constants) what one kilogram is [157, 158].

So, since there is indeed a proportionality relation between the energy and frequency of a wave, if Maria "taps into this energy" (the how being that Maria is able to get the frequency of anything, then her mind can multiply this frequency by the well known Plank constant to obtain the corresponding energy), then she'll be able to "see glimpses" of her friends. Right?

Well, as Rosa says, there indeed was a CIA program, Remove Viewing [159], which is not backed by any science, lacks control and repeatability of the experiments, and failed to produce any sound intelligence information.

So, whether Maria can or cannot do this is of the realm of fiction, more so than of science.

[55] I did try to keep it brief, but it is still quite long.

However, that being said, let's look at whether this is a sound scientific reasoning or not!

First, two things that Maria can do once she has "taped into the energy". One, she "sees" *them* in a vision. Two, she "sees" their *location*. These are similar concepts, but what I mean in the latter is that she knows where they are, like if you share your location with friends you see a dot on a map of where they are. The former concept as no scientific proof, so I am ignoring this one.

This latter concept is not entirely backed up by science. The only somewhat correlation between distance and frequency there is (as far as I'm aware) is the *(relativistic) Doppler effect* [160, 161]. However, this effect relates the frequency and the velocity of a moving source with regards to a fixed observer (and vice versa). So, for this to work, Maria would need to stay fixed in space, and the friends would need to move. Then, based on the change of frequency, she might be able to determine the speed at which they are moving. From this, and from the frequency and time (which are related), she might be able to figure out the distance they traveled. But that is quite vague as to the distance at which they are from her.

From the Doppler effect, Maria would be able to determine in which direction (away or towards) her the friends are traveling at. This can be found based on the whether the frequency increases (the source is traveling towards the observer) or decreases (the source is traveling away).

So, Maria can determine the speed at which the friends are travelling and the direction (away or towards her).

To get an indication of the distance at which they are from her, she would need more information, which is not straightforward to obtain. She would need to know the time it takes for a wave emitted by the source (friends) to reach her (the observer). So, she would need to have a clock at the source, or in other words, a stopwatch that starts when the wave is emitted and stops when it reaches her. Technically feasible since there is a way to communicate as shown by Alex is previous episodes, but it would require a controlled environment, and since Alex was communication by moving things around, then Maria would already be able to determine where the friends are.

To conclude on this, with the frequency of the wave, Maria can determine quite a lot about the whereabouts of her friends, so adding to this the psychic element, it is quite logical that she would be able to determine where they are from this.

As a side note, I personally prefer the aspect of seeing *where they are* versus seeing *them*. I think it's quite fun to imagine Maria having a map of Roswell in her mind, and having the little dot moving around representing where her friends are. And less invasive of everyone's privacy.

The next bit of science is from Liz. Max interrupted her, and she starts listing what she knows.

> *"Do you understand the relativistic dynamics of Hawking's radiation? Can you convert constrained Lagrange multipliers?"*

Breaking up all of this into parts, we have

- Relativistic dynamics of Hawking's radiation.

DEFINITION 4.10.3: HAWKING RADIATION

Hawking radiation was developed in Stephen Hawking in 1974 and is a model that includes radiation outside of a black hole. It is an extremely faint radiation, or many orders of magnitude lower than what can be detected by telescopes today [162].

DEFINITION 4.10.4: RELATIVISTIC DYNAMICS

Relativistic dynamics refers to the relation between motion and properties of a relativist system, i.e. in special relativity, the theory of space and time [163, 164].

These are all concepts that do exist, but Liz is supposed to be a biologist who specializes in stem cell research. I understand having hobbies in different branches of science, but even as someone who took a couple astronomy courses rather recently, I wouldn't necessarily say that I understand the relativistic dynamics of Hawking's radiation.

- Convert constrained Lagrange multipliers[56].

DEFINITION 4.10.5: LAGRANGIAN FUNCTION AND MULTIPLIER

A **Lagrangian function**, or simply a Langrangian, is a function defined as

$$\mathscr{L}(x,\lambda) = f(x) + \langle \lambda, g(x) \rangle,$$

where $\langle , \rangle$ denotes an inner product, f and g are functions, and λ is the **Lagrangian multiplier** [165].

DEFINITION 4.10.6: INNER PRODUCT

An **inner product**, is a map which is positive definite, linear and symmetric under conjugation. It produces a scalar [166].

For example, consider the real numbers R (any number you can think of, basically), then the inner product is simply the multiplication $\langle x,y \rangle = xy$. For a vector space $\mathbb{R}^n$, the inner product is the dot product $\langle x,y \rangle = x \cdot y = x^T y$ (where x^T denotes transposition). Illustrating,

$$\langle 2,3 \rangle = 2 \times 3 = 6$$
$$\left\langle \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \right\rangle = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = 1 \times 4 + 2 \times 5 + 3 \times 6 = 32$$

[56] Heads up, this is a mathematical concept, and I love maths. I will try my best to keep this short, concise, and understandable.

Lagrangian functions are used in the field of mathematical optimization, with the aim to find local minima and maxima of a function subject to some constraints. In the above definition of $\mathscr{L}$, f is the function that we are trying to find the minima or maxima of, and g is the constraint.

The word constrained in Liz's sentence is a little redundant, since by default we use Lagrangians when there is a constraint. I am not entirely sure about what she means by "convert". The only conversion I can think of is between the initial minimisation (or maximisation) constrained problem and the Lagrangian. For instance, the following minimisation problem can be written equivalently in terms of the Lagrange multiplier as

$$\begin{aligned} &\text{Minimise } f(x,y) \text{ subject to } g(x,y) = c \\ \iff &\text{Find the points } (x,y) \text{ such that } \nabla f(x,y) = \lambda \nabla g(x,y), \end{aligned}$$

where ∇ denotes the gradient operator [167].

This does have applications in biology, for instance to determine polymer properties subject to constraints. Again, I am not a biologist, but I did find two papers that mention Lagrangians in biology and chemistry [168, 169]. However, I do think some of the words in this sentence do not add any meaning, and if anything, are only more confusing.

In conclusion, what Liz mentions are indeed real life science. However, I do feel like they were only added as technobabble, and the writers could have referred to some biological concepts closer to what Liz's research is about.

The next bit of science already gives me a headache before even starting. It's between Liz and Shivani. **Trigger warning for mentions of animal testing, blood, dead beings, and draining**[57].

Liz: *"This frog's nucleating proteins ensure the water in its blood freezes first. So the cells shrink, and it enters suspended animation."*

Shivani: *"An earthbound way of replicating what those alien pods do."*

Liz: *"But our Kermit here is freshly dead, so I have an electrical current that's pulsing through him. Same as you have to with Nicole. And it's gonna stimulate and simulate stasis. But with the right chemical combo, we'll breathe new life into those waiting cells."*

A lot of words. Let's break them up.

DEFINITION 4.10.7: NUCLEATION

Nucleation is a process that initiates phase transition, via self-organisation of a substance. Self-organisation here means that the molecules will arrange themselves in an orderly manner, chang-

[57] ?? at this point I don't know either.

ing the structure of the substance. Nucleation of proteins also leads to the formation of a variety of states and phases. Namely, one can reach crystalisation of proteins, which has many pharmaceutical applications [170, 171, 172].

Nucleation is also the first step of freezing in cryogenisation [173]. Species that are able to survive freezing also use ice nucleation, and this is quite a known process [174].

There are two kinds of ice nucleation (nucleation leading to freezing); homogeneous in which ice forms without the presence of foreign particles, and heterogeneous in which foreign molecules are introduced to the substance which leads to ice formation.

Fun fact on the topic of frogs, there is a species of frogs called wood frogs which freezes in the winter, and becomes completely solid! It's brain and heart stop working, and in spring the frog starts to thaw, and it's organs regain their normal functions. It's blood and skin tissue are also frozen during this process [175].

With regards to the water in the blood freezing first, the freezing point for water is 0°C (=32°F) and for human blood it is about -2°C or -3°C (=28.4°F and 26.6°F respectively) [176]. Note that blood can only freeze outside of a (live) body, since the metabolism of the body will naturally keep it sufficiently warm.

So, as far as my research went, I don't think there is anything outrageous about what Liz says.

Side note on Nicole being frozen. A quick google search has lead me to learn that in cryonics, blood is removed from the dead bodies, and replaced with antifreeze and organ preserving compounds, called cryoprotective agents [177]. This has made me realise that Nicole would be preserved in the cryogenic pod without blood, and instead with another liquid which would preserve her organs better. In our world, the only bodies that end up frozen are not alive anymore, so this is not that accurate. Moreover, it appears that freezing a live body is a fatal process, with very low chance of being reversed [178]. So, unless she became a wood frog, this whole plot is definitely fiction.

The last part of what Liz says is also, as far as I'm aware, fiction. Since, again, human's are not wood frogs, once we are frozen, bringing life back is not (yet) possible. However, in the realm of fiction and aliens, the pod that Nicole is in mimicks the alien pods, and thus injecting some alien compound (literally) would in a way defreeze her organs, stop the stasis, and eventually bring her (and the frog) back to life.

The scene continues with the frog being injected with sparkly alien substance and being brought back to life. Then, Shivani immediately jumps to the next step: her daughter.

I have many things to say about this scene, but I will keep it short. This is not how a scientific experiment is held.

First, there needs to be more trials held, with more frogs, and then progressively with mammals like mice. Then, bigger mammals. Finally, humans. Again, we are not frogs. An important aspect of science is the reproducibility of results. Results obtained once are not sufficient, we need to obtain the same results all the time, under different varying parameters and conditions.

At least Liz is also of the opinion that having a frog brought back to life is not enough to directly jump to human trials. Later in the episode the frog dies again. Proof by example of why more trials are necessary, with more tests about the trials.

4.11 FOLLOW YOU DOWN

The first bit of science consists in Liz writing on a clear board. I went back to watch the scene, but I unfortunately couldn't get a clear view of what she was writing. It's almost like the writers didn't want me to be able to analyse whether what she was writing is correct or not. Then, she says:

> *"I can't remember the genome sequence to rapid amplification..."*

Recall that the genome (Definition 2.4.1, Definition 2.4.2) is the complete set of genetic information in an organism. Genome sequencing is the process of determining the entire DNA sequence of a organism's genome.

DEFINITION 4.11.1: RAPID AMPLIFICATION

Rapid amplification is a technique used in molecular biology to obtain the full sequence of an RNA transcript found within a cell [179].

Recall that RNA (Definition 3.1.2) is similar to DNA, but it is single-stranded.

The words genome sequence to rapid amplification don't really seem to mean anything, or at least not how they are written here. Particularly, rapid amplification seems to be related to RNA while genome sequencing to DNA, which are not quite the same thing. However, rapid genome amplification is a thing that exists, it is a faster method to obtain many copies of an entire genome [180].

Let's just say Liz is confused because of having inhaled the mist.

In the same scene, Kyle seems to also be worried about Liz's state:

> *"I'm working on a hybrid methadone that'll help ease the withdrawal symptoms."*

DEFINITION 4.11.2: METHADONE

Methadone is a synethic opioid that is used to treat chronic pain and opioid addiction [181].

I guess his running theory here is that Liz is showing signs of opioid addiction, meaning that the mist that she inhaled would appear to share some characteristics with opiods.

In the same scene again, Max asks a very good question[58]:

Max: "What are these equations, Liz?"

Liz: "I am charting the nucleotides in your [Max's] triple helix to see if I can fight fire with actual fire."

Nucleotides (Definition 2.4.3) are at the base of all hereditary characteristics.

The triple helix mentioned here refers to the fact that the oasians have one more helix than humans. Hence, here Liz is analysing the specific alien component of Max's DNA, and looking at his inherited characteristics.

The next piece of science is particularly interested to me for two aspects.

Dallas: "Wait, these old shipping containers are coated in lead. That means electromagnetic waves can't pass through them."

Bonnie: "What does that mean?"

Dallas: "When I first got inducted into the pod squad, Michael told me that our powers are electromagnetic."

The first aspect is that Alex build a Faraday cage, knowing that it would block the oasian powers (very cool). The second aspect that I find very interesting is that Bonnie that doesn't seem to know that their powers have an electromagnetic base. (Alternatively, the writers could have made Bonnie ask this question to have Dallas explain what he means, but I prefer a Watsonian explanation). Hence, it could be that on Oasis, their powers and how they work are not necessarily studied by all, which would make sense, as we humans don't all know how our bodies work.

I will not be detailing too much on electromagnetic waves (I have already written quite a lot on this before), but I will go in details on how a Faraday cage works (just because I find this really interesting).

DEFINITION 4.11.3: FARADAY CAGE

A **Faraday cage** works as a shied to block electromagnetic waves from passing through. They are named after Michael Faraday, who first built one in 1836. They block electromagnetic waves for which the wavelength is larger than the holes of the cage [182].

Basically, when an electromagnetic wave with wavelength larger than the holes of the cages hits it, the electric charges will disperse on the cage and cancel out the effect of electric field inside of the cage. And hence, it doesn't enter the cage (or doesn't exit it, depending on the point of view, and here Bonnie and Dallas are inside of the cage, but their powers cannot go through it).

The best example of a Faraday cage is a car or a plane. If you are in your care during a lightning storm, you should stay inside of it! If lightning hits your car, and you are inside, and **you are not touching any of the**

[58] Mostly because I am wondering the same thing since *I cannot read the board!*

walls of your car you will be safe. If, however, you have e.g. your hand on the door, then the electric current of the lightning will transfer through contact to you, and you will get electrocuted.

The next science bit in the episode is also an interesting philosophical question.

Shivani: "Real life? Which is governed by physics, chemistry, biology. Equal and opposite reactions. Cause and effect. Our entire lives written into our very DNA before we even take a first step. One big chain reaction leading to one inevitable conclusion."

Shivani carries the idea that our DNA predicts our future, every decision we make. That before we are even born, as our DNA forms as we are fœtuses, our destiny is already written.

There is an interesting Scientific American blog post about this [183], which mentions books published on how DNA acts as a 'fortune-teller'. Moreover, an article published on the BBC in 2023 [184] mentions that the '*complex cocktail of DNA we inherit from our parents, along with around 70 spontaneous genetic mutations which we acquire by chance, subconsciously dictates our behaviour to a far greater extent than we are aware.*'

On the other hands, many scientists believe that our future is not determined by our DNA from the moment we are born, but that our environment changes our gene expressions, leading to a strong link between our decisions and our genes. This is the field of **epigenetics** [185]. Note however, that in this field, it is not necessarily considered that the DNA changes.

This is a complex debate, and one that is still happening, and ever evolving as scientists learn more about our DNA. A blog post from the Max Planck institute [186] talks about the nuances in this debate, and how both the DNA and the environment are found to contribute to who we are.

I will not delve further into this conversation, I just think it's an interesting development of science, and I do appreciate that it was mentioned by Shivani. Especially since in the show, we do have examples of cases where DNA is not the only factor in determining who someone is (for instance, Max being a carbon copy of Jones, but not a genocidal maniac).

We then see Liz holding a piece of paper with equations written by Theo. I would love to rewrite what these equations say, however, it is very hard to decipher what is actually written. Moreover, it's equations about removing the triad tatoo from Clyde, so even if I'm able to decipher what is written, I doubt it'll make much sense to me.

Liz then says some very simple and very true science!

"I need ferric oxide, i.e. rust."

DEFINITION 4.11.4: FERRIC OXIDE

Ferric oxide is red iron oxide, which is often called rust. However, scientifically speaking, rust is not a well-defined concept [187]

The next science mention is (again) on the topic of frequencies. I have already written something like 3 thousand words on frequencies before, so I will try to keep this brief.

Rosa: "Liz figured out that resonate frequencies can break psychic tethers and, with four roommates, I've gotten kind of good at silencing them, so maybe."

Rosa: "The frequency's destabilizing."

The first part of what Rosa says – resonate frequencies can break psychic tethers – relates to what Liz says (and does) in episode 3x09 (see Section 3.9). As I said in my meta for that episode, it is alien science based on some real science (the alien part behind the psychic element here).

The following science element is Alex and the consequence of the pocket dimension on his well being. He says:

Alex: "If this is an alternate dimension, the amount of energy that it would take to build it would border on atomic fission. And like nuclear power, it would have unintended consequences. Radiation."

Michael: "The gas mask."

Alex: "I felt nauseous as soon as I got here. Did you? Dallas, Bonnie? [...] Okay, so it's only humans. Which is why Theo didn't realize. [...] The gas mask helped with the nausea, but it-it couldn't stop this. [...] I think I'm dying."

He also shows Michael black fractal burns that have spread on his chest, see Figure 4.4.

The first part of what Alex says would make sense to me, that the pocket dimension would require a huge amount of energy to be built and to not collapse on itself.

Some people have made the calculations for the amount of energy needed to build a pocket dimension (see this blog post), where it amounts to about $10^4 1$ J. This is indeed a lot of energy. For comparison, a lethal dose of radiation absorbed is about 4.4 Gray = 4.4 J/kg (Joule per kilogram). Suppose Alex weighs 80 kg (average weight in North America, 177 lbs), then the lethal dose would be 352 J. (Which would mean that he would likely be already dead, so let's assume that this specific pocket dimension took a lot less energy to be created).

Spefically looking at Alex's burns (Figure 4.4), in the shape of fractals, this resembles more lightning scars than radiation burns. I will let you look at google images to see this for yourself. I will just conclude by saying that Alex's burns like this because this is oasian radiation, where the fission is with different elements

Figure 4.4. Screenshot in black and white showing the black fractals on Alex's chest due to the radiation from the pocket dimension[59].

than the standard nuclear fission on Earth (in my opinion, this would be the only factor that would make the burns look different).

The final bit of science in this episode is Liz being a total badass once again!

Liz: "You were right not to trust me. The mist was synthesized from Bonnie's virus, and by reverse engineering the chemical structure, I was able to mimic her abilities and render you powerless."

I will not comment on this too much since it's alien science and therefore out of the scope of my meta. But basically, Liz sampled the virus in the mist, where instead of using it to give powers (the initial reverse engineering made to create the mist and give Liz her powers), she reversed-reversed engineered (or in other words, recovered the initial virus), which she then injected Clyde with. Thus, infecting him with Bonnie's virus.

After a quick search, I did find that it is possible to recover bacteria from antibiotics, so what Liz is doing does seem to have roots in human science [188].

4.12 TWO SPARROWS IN A HURRICANE

The first bit of science in this episode is very much alien science, where Liz is working to get Max's powers back. Max had gone to Senovative (Shivani's company), only to find that it was deserted. He shows Liz a mostly erased whiteboard, and she says that she doesn't recognise the equation. I looked at said equation and I honestly can't read anything off of it. But considering it's what Shivani and Clyde were working on, I'm seriously doubting I'd understand it either way.

The next science bit is between Liz and Kyle, where they are discussing the effect of the mist. Kyle is writing on a see through board (which I once again cannot really decipher). This is all real-life science.

Liz: "'C7H15Cl.' What is that?"

Kyle: "Cyclophosphamide? The alkylating agent used in cancer treatments."

Liz: "Yeah. Methadone nap must've given me a brain cramp."

First, Liz mentions a molecular formula. It corresponds to the following compound:

DEFINITION 4.12.1: 1-CHLOROHEPTANE

1-Chloroheptane is a chemical compound with the molecular formula $C_7H_{15}Cl$ [189]. It is a colourless iquid which is insoluble in water, both flammable and irritant.

It is a *halogenated alkane* and serves as a valuable solvent in various chemical applications, where its primary use is in *chromatography* and *spectroscopy*, namely for the analysis of halogenated organic compounds. For example, it can be used as a calibration standard [190].

DEFINITION 4.12.2: CHROMATOGRAPHY & SPECTROSCOPY

Chromatography is a technique for the separation for a mixture into its components [191].
Spectroscopy is the study of electromagnetic spectrum as it interacts with matter [192].

DEFINITION 4.12.3: HALOGEN

Halogenation is a chemical reaction which introduces some *halogens* into a chemical compound, used e.g. to produce *polymers* or drugs [193].
Halogen is a group of the periodic tables consisting of: fluorine (F), chlorine (Cl), bromine (Br), iodine (I), astatine (At) and tennessine (Ts). The latter two are radioactive elements [194].

Recall that a polymer is a material consisting of large molecules (Definition 3.7.1).

DEFINITION 4.12.4: ALKANE

In chemistry, an **alkane** is an organic compound consisting of hydrogen and carbon, with the general chemical formula C_nH_{2n+2} for n arbitrary integer. The simplest alkane is methane for $n = 1$ of CH_4 [195].

Hence, a **halogenated alkane** is an alkane in which a halogen was introduced. For instance, **haloalkanes** are alkanes which contain one or more halogen as substitutes for hydrogen [196].

Kyle then answers by mentioning the name of a medication:

DEFINITION 4.12.5: CYCLOPHOSPHAMIDE

Cyclophosphamide, aslo known as **cytophosphane**, is a medication used in chemotherapy to suppress the immune system. It is used for isntance to treat mulitple forms of cancer. Its chemical formula is $C_7H_{15}Cl_2N_2O_2P$ [197].

So indeed the start of it is what Liz mentioned. As Kyle was writing on the board while she asked her first question, it is probable that Kyle hadn't finished writing the formula entirerly.

The rest of Kyle's sentence is also what is written on the Wikipedia page for Cyclophosphamide. Clearly, as Liz says in the next sentence, she should have known this, and is still suffering from effects from having inhaled the mist.

Liz's mention of methadone refers back to the previous episode, where Kyle gave her a methadone to fight the withdrawal symptoms of the mist.

In the same scene, Rosa arrives and talks to Liz as her sponsor, showing concern about there being more mist present near Liz. Liz reassures her by saying

> *"Look, Kyle ran me through the elevated risk of stroke and myocarditis, and since nothing speaks to me like the science, we're safe. No more mist."*

As a side note, I also agree with Liz to some extent. The best way for to understand the risks of dangers of something is by learning the science behind it.

DEFINITION 4.12.6: MYOCARDITIS

Myocarditis is an inflammation of the heart muscle that can lead to a heart attack. It is most often due to infection, but can also be caused by some medication, toxins or autoimmune disorders [198].

From the list of causes on Wikipedia, there wasn't anything that really jumped to me as being related to the mist. But there are some toxins and stimulants listed there, so it could be that one of those is also present in the mist.

The rest of the 'science' in this episode is alien, from the stuff with the pocket dimension and travelling to it, to getting out of it, to Bonnie's DNA containing bits of the DNA of everyone she's ever kissed...

This also marks the end of the science analysis for season 4 (and with it for the series), since in episode 4x13 the only mention of real-life science is more on frequencies. I believe that I've already talked enough about frequencies here.

CONCLUSION

This marks the end of my episode-by-episode breakdown of the science in the TV Show Roswell, New Mexico. This journey took four years to complete. I started it for the #rnmmarchformeta2022 event, as I was merely a second year bachelor student in applied mathematics. I hadn't yet written a thesis, I didn't know much about proper scientific writing or research, but I was passionate about both this show and science in general. I wanted to combine both interests, and I wanted to make the science babble as understandable as possible. I hope that I succeeded in that.

As I am writing these concluding words, I have completed a master's degree in applied mathematics, and I am about to begin a PhD in fluid dynamics. I have written three theses, and many more reports, and I aim to make scientific research my future job.

This meta is longer than all of my theses. It is over 30 thousand words, while my theses are respectively 17 thousand, 5 thousand and 11 thousand words. Roughly speaking, this meta is as long as all my theses, and took longer to write. It is also on a topic that I am significantly less familiar than my major.

Some more statistics on this meta. I have collected all the dialogues that I have analysed, and I have collected the number of times each character has a line of dialogue containing some scientific notion in Table 4.1. Not all characters have been listed, only those with more than 2 lines of dialogue were considered, so that the table is not too large.

	Liz	Kyle	Alex	Heath	Shivani	Diego	Michael	Rosa	Dallas	Isobel	
S1	12	3	2	0	0	0	0	0	0	0	**17**
S2	10	1	0	0	0	5	1	0	0	0	**17**
S3	8	1	1	7	0	0	2	0	0	1	**20**
S4	33	5	6	2	7	0	2	4	3	1	**63**
	63	**10**	**9**	**9**	**7**	**5**	**5**	**4**	**3**	**2**	**117**

Table 4.1. Number of lines of dialogue containing scientific terms per character per season. The bold numbers are the sum of each row/column.

Evidently, and I don't think you need a table to know this, Liz has the most technobabble of all, by far. Second and third are Kyle and Alex, respectively. Then, the next three are very interesting to me. Heath, Shivani, Diego. The latter two have given me multiple headaches over the course of seasons 2 and 4 (for Diego and Shivani respectively). I am surprised that Michael is not higher in the list, but I guess his scientific contribution is with his hands and not by his dialogue.

What can also be seen in this table is the distribution of science-related dialogue over the seasons. Season 4 has clearly the most, while Seasons 1-3 are closer in numbers. This could be explained by two main reasons.

- The last season has more elements that need to be explained in words, namely a deeper exploration of the alien powers, the pocket dimension, everything to do with Shivani and her daughter being in a cryogenic pod, to name a few. This season also contains more or less one entire episode dedicated to St Elmo's fire, and the weather phenomenon has repercussions over multiple episodes. Liz also inhales the mist with the aim of being a better scientist, thus increasing her technobabble. However, this also causes her to have memory issues, making her ask more questions (especially towards the last episodes).
- I have analysed more dialogue towards the end of the show than at the start. For context, Seasons 1 through 3 were analysed over the course of a couple of months only, and more importantly, I discussed them in one go. Meaning, I rewatched all three seasons (and read the transcripts) in a short amount of time. This short time has most likely lead me to skip a few things, to maybe decide to not analyse a certain dialogue as it perhaps sounded too alien-y for this meta. On the other hand, for Season 4, I began by analysing each episode in the week between two episodes airing. Therefore, more time was spent on each episode, allowing for a deeper analysis on my end.

I do think that both these facts can be simultaneously true. However, I do take responsibility in my choice of what to analyse and what to skip. To this end, it is entirely possible than this meta gets completed in the future, adding more explanations to some of the technobabble that was initially skipped.

Finally, I would like to thank everyone who encouraged me in the four years that it took me to write this meta. For the comments I got both on Tumblr or Discord, for the support and the brainstorming sessions, thank you.

It has truly been my pleasure to write this, and I hope that it can help someone understand what was said in this show a little better.

REFERENCES

[1] Wikipedia contributors. *Phenylacetone — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Phenylacetone&oldid=1150669482. 2023.

[2] National Center for Biotechnology Information. *PubChem Compound Summary for CID 7678, Phenylacetone*. https://pubchem.ncbi.nlm.nih.gov/compound/Phenylacetone.. 2023.

[3] Wikipedia contributors. *Amphetamine — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Amphetamine&oldid=1149937750. [Online; accessed 27-April-2023]. 2023.

[4] Wikipedia contributors. *Methamphetamine — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Methamphetamine&oldid=1150930763. 2023.

[5] L. NJ, H. AL, and C. JP. "Stem cell therapy for vascular regeneration: adult, embryonic, and induced pluripotent stem cells". *Circulation* (2010). https://doi.org/10.1161/CIRCULATIONAHA.109.881441.

[6] Wikipedia contributors. *Hypothalamic–pituitary–adrenal axis — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/Hypothalamic-pituitary-adrenal_axis. 2026.

[7] Wikipedia contributors. *Hypothalamus — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/Hypothalamus. 2026.

[8] Wikipedia contributors. *Pituitary gland — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/Pituitary_gland. 2026.

[9] Wikipedia contributors. *Adrenal gland — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/Adrenal_gland. 2026.

[10] Wikipedia contributors. *Angiogenesis — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Angiogenesis&oldid=1144082867. 2023.

[11] Wikipedia contributors. *Transcutaneous pacing — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Transcutaneous_pacing&oldid=1128400918. 2022.

[12] National Institutes of Health. *Stem Cell Basics*. https://stemcells.nih.gov/info/basics/stc-basics. 2021.

[13] Wikipedia contributors. *Laparotomy — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Laparotomy&oldid=1133657198. 2023.

[14] Wikipedia contributors. *Epithelium — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Epithelium&oldid=1147526984. 2023.

[15] R. K. Josephson and W. E. Schwab. "Electrical properties of an excitable epithelium." *Journal of General Physiology* 74.2 (1979). https://doi.org/10.1085/jgp.74.2.213, pp. 213–236.

[16] J. Moleón Baca et al. "Cells electric charge analyses define specific properties for cancer cells activity". *Bioelectrochemistry* 144 (2022). https://doi.org/10.1016/j.bioelechem.2021.108028.

[17] J. Pu, L. Cao, and C. D. McCaig. "Physiological extracellular electrical signals guide and orient the polarity of gut epithelial cells". *Tissue Barriers* 3.3 (2015). https://doi.org/10.1080/21688370.2015.1037417, e1037417.

[18] Wikipedia contributors. *Coronary arteries — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Coronary_arteries&oldid=1139946975. 2023.

[19] Wikipedia contributors. *Cytoplasm — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Cytoplasm&oldid=1151838817. 2023.

[20] Wikipedia contributors. *Enzyme — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Enzyme&oldid=1132111677. 2023.

[21] Britannica. *Protein*. https://www.britannica.com/science/protein.

[22] Wikipedia contributors. *Catalysis — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Catalysis&oldid=1151111680. 2023.

[23] Wikipedia contributors. *Exothermic process — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Exothermic_process&oldid=1136634491. 2023.

[24] Wikipedia contributors. *Fibonacci sequence — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Fibonacci_sequence&oldid=1151220286. 2023.

[25] Wikipedia contributors. *Golden ratio — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Golden_ratio&oldid=1150269490. 2023.

[26] Wikipedia contributors. *Fibonacci search technique — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Fibonacci_search_technique&oldid=1136333815. 2023.

[27] Geek for Geeks. *Fibonacci Search*. https://www.geeksforgeeks.org/fibonacci-search/. 2023.

[28] Wikipedia contributors. *Shor's algorithm — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Shor%27s_algorithm&oldid=1150889930. 2023.

[29] Geek for Geeks. *Shor's Factorization Algorithm*. https://www.geeksforgeeks.org/shors-factorization-algorithm/. 2023.

[30] Wikipedia contributors. *Phosphorus — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Phosphorus&oldid=1150855572. 2023.

[31] S. Kwon, Y. Kwon, T. Lee, et al. "Recent advances in stem cell therapeutics and tissue engineering strategies". *Biomater Res* 22 (2018). https://doi.org/10.1186/s40824-018-0148-4.

[32] S. Diederichs, K. M. Shine, and R. S. Tuan. "The promise and challenges of stem cell-based therapies for skeletal diseases". *BioEssays* 35 (2013). https://doi.org/10.1002/bies.201200068.

[33] J. Cieśla and M. Tomsia. "Cadaveric Stem Cells: Their Research Potential and Limitations". *Frontiers Genetics* (2021). https://doi.org/10.3389/fgene.2021.798161.

[34] H. Shenghui, D. Nakada, and S. J. Morrison. "Mechanisms of Stem Cell Self-Renewal". *Annual Review of Cell and Developmental Biology* 25.1 (2009). https://doi.org/10.1146/annurev.cellbio.042308.113248, pp. 377–406.

[35] B. Bussolati. "Stem cells for organ repair". *Organogenesis* 7.2 (2011). https://doi.org/10.4161/org.7.2.16605, pp. 95–95.

[36] S. Montagnani et al. "Adult Stem Cells in Tissue Maintenance and Regeneration". *Stem Cells International* (2016). https://doi.org/10.1155/2016/7362879.

[37] University of Leiden. *Curing diseases with lab-grown organs*. https://www.universiteitleiden.nl/en/news/2022/11/curing-diseases-with-lab-grown-organs. 2022.

[38] Wikipedia contributors. *DNA sequencer — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=DNA_sequencer&oldid=1144105105. 2023.

[39] Wikipedia contributors. *DNA sequencing — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=DNA_sequencing&oldid=1150748780. 2023.

[40] Wikipedia contributors. *Genome — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Genome&oldid=1151521799. 2023.

[41] Britannica. *Nucleotide*. https://www.britannica.com/science/nucleotide. 2026.

[42] Wikipedia contributors. *Apoptosis — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Apoptosis&oldid=1150813710. 2023.

[43] Wikipedia contributors. *Polymerase — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Polymerase&oldid=1075128624. 2022.

[44] Wikipedia contributors. *Biopolymer — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Biopolymer&oldid=1146139576. 2023.

[45] B. Carneiro and W. El-Deiry. "Targeting apoptosis in cancer therapy". *Nature Reviews Clinical Oncology* 17 (2020). https://doi.org/10.1038/s41571-020-0341-y, pp. 395–417.

[46] Wikipedia contributors. *Craniopagus twins — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Craniopagus_twins&oldid=1148096665. 2023.

[47] Wikipedia contributors. *Antibody — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Antibody&oldid=1149824703. 2023.

[48] Wikipedia contributors. *Propranolol — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Propranolol&oldid=1148773447. 2023.

[49] Wikipedia contributors. *Norepinephrine — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Norepinephrine&oldid=1150130003. 2023.

[50] Wikipedia contributors. *Neurotransmitter — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Neurotransmitter&oldid=1150258312. 2023.

[51] C. Young and R. Butcher. *Propranolol for Post-Traumatic Stress Disorder: A Review of Clinical Effectiveness*. https://www.ncbi.nlm.nih.gov/books/NBK562942/. Canadian Agency for Drugs and Technologies in Health, 2020.

[52] Wikipedia contributors. *Butyric acid — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Butyric_acid&oldid=1151927210. 2023.

[53] A. Belén Sanz-Martos et al. "Butyric Acid Precursor Tributyrin Modulates Hippocampal Synaptic Plasticity and Prevents Spatial Memory Deficits: Role of PPARγ and AMPK". *International Journal of Neuropsychopharmacology* 25.6 (2022). https://doi.org/10.1093/ijnp/pyac015, pp. 498–511.

[54] Wikipedia contributors. *Neoblast — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Neoblast&oldid=1007564718. 2021.

[55] Wikipedia contributors. *Cellular differentiation — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Cellular_differentiation&oldid=1151646842. 2023.

[56] Wikipedia contributors. *Flatworm — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Flatworm&oldid=1150028000. 2023.

[57] Wikipedia contributors. *Bilateria — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Bilateria&oldid=1142409807. 2023.

[58] E. Tsarkova et al. "A Study on the Planarian Model Confirms the Antioxidant Properties of Tameron against X-ray- and Menadione-Induced Oxidative Stress". *Antioxidants* 12.4 (2023). https://www.mdpi.com/2076-3921/12/4/953.

[59] W. Zakrzewski et al. "Stem cells: past, present, and future". *Stem Cell Research & Therapy* 10 (68 2019). https://doi.org/10.1186/s13287-019-1165-5.

[60] R. L., I. P., and S. A. "Stem cells and neural signalling: the case of neoblast recruitment and plasticity in low dose X-ray treated planarians". *The International Journal of Developmental Biology* 56 (2012). https://doi.org/10.1387/ijdb.123505lr, pp. 135–142.

[61] M. Webster. *Inducible*. https://www.merriam-webster.com/dictionary/inducible. 2023.

[62] Wikipedia contributors. *Diphtheria — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Diphtheria&oldid=1147864520. 2023.

[63] C. E. Canal, Q. Chang, and P. E. Gold. "Amnesia produced by altered release of neurotransmitters after intraamygdala injections of a protein synthesis inhibitor". *Proceedings of the National Academy of Sciences* 104.30 (2007). https://www.pnas.org/doi/abs/10.1073/pnas.0705195104, pp. 12500–12505.

[64] P. E. Gold. "Protein synthesis inhibition and memory: Formation vs amnesia". *Neurobiology of Learning and Memory* 89.3 (2008). https://doi.org/10.1016/j.nlm.2007.10.006, pp. 201–211.

[65] Wikipedia contributors. *Γ-Aminobutyric acid — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=%CE%93-Aminobutyric_acid&oldid=1151743311. 2023.

[66] Wikipedia contributors. *Dorsolateral prefrontal cortex — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Dorsolateral_prefrontal_cortex&oldid=1137777442. 2023.

[67] T. J. Legg. *Can Depression Cause Memory Loss?* https://www.healthline.com/health/depression/depression-and-memory-loss. 2016.

[68] Wikipedia contributors. *Calcium carbonate — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Calcium_carbonate&oldid=1150823706. 2023.

[69] GenScript. *Excision*. https://www.genscript.com/biology-glossary/1002/excision. 2023.

[70] Wikipedia contributors. *Nucleobase — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Nucleobase&oldid=1147023086. 2023.

[71] Wikipedia contributors. *RNA — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=RNA&oldid=1150499453. 2023.

[72] Wikipedia contributors. *Plexus — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Plexus&oldid=1140100073. 2023.

[73] Wikipedia contributors. *Cerebrospinal fluid — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Cerebrospinal_fluid&oldid=1147468913. 2023.

[74] Wikipedia contributors. *Plasmid — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Plasmid&oldid=1151843704. 2023.

[75] Wikipedia contributors. *Endospore — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Endospore&oldid=1144280073. 2023.

[76] Wikipedia contributors. *Lidocaine — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Lidocaine&oldid=1147093408. 2023.

[77] Alzheimer's Association. *Vascular Dementia*. https://www.alz.org/alzheimers-dementia/what-is-dementia/types-of-dementia/vascular-dementia. 2023.

[78] C. of Disease Control and Prevention. *African Trypanosomiasis*. https://www.cdc.gov/parasites/sleepingsickness/index.html. 2022.

[79] Wikipedia contributors. *African trypanosomiasis — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=African_trypanosomiasis&oldid=1148989176. 2023.

[80] Wikipedia contributors. *Laparoscopy — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Laparoscopy&oldid=1151017181. 2023.

[81] Wikipedia contributors. *Polymer — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Polymer&oldid=1149449548. 2023.

[82] Wikipedia contributors. *Hydrolysis — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Hydrolysis&oldid=1148947066. 2023.

[83] Wikipedia contributors. *Hertz — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Hertz&oldid=1141628634. 2023.

[84] Wikipedia contributors. *Fundamental frequency — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Fundamental_frequency&oldid=1148631124. 2023.

[85] Wikipedia contributors. *Waveform — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Waveform&oldid=1151442426. 2023.

[86] Wikipedia contributors. *Beta wave — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Beta_wave&oldid=1094097353. 2022.

[87] B. Kenkel. *Nanoblades: Tiny CRISPR Ninjas for Genome Editing Difficult Cells*. https://blog.addgene.org/nanoblades-crispr-genome-editing-difficult-cells. 2019.

[88] D. P. Perl. "Neuropathology of Alzheimer's Disease". *Mount Sinai Journal of Medicine: A Journal of Translational and Personalized Medicine* 77.1 (2010). https://onlinelibrary.wiley.com/doi/abs/10.1002/msj.20157, pp. 32–42.

[89] Wikipedia contributors. *Endoplasmic reticulum — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Endoplasmic_reticulum&oldid=1151735252. 2023.

[90] Wikipedia contributors. *Eukaryote — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Eukaryote&oldid=1151020508. 2023.

[91] Wikipedia contributors. *Cipher — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Cipher&oldid=1144227515. 2023.

[92] Wikipedia contributors. *Vigenère cipher — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Vigen%C3%A8re_cipher&oldid=1145986634. 2023.

[93] Geek for Geeks. *Vigenère Cipher*. https://www.geeksforgeeks.org/vigenere-cipher/. 2023.

[94] Wikipedia contributors. *Caesar cipher — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Caesar_cipher&oldid=1151194063. 2023.

[95] Geek for Geeks. *Caesar Cipher in Cryptography*. https://www.geeksforgeeks.org/caesar-cipher-in-cryptography/. 2023.

[96] Wikipedia contributors. *Autokey cipher — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Autokey_cipher&oldid=1151279094. 2023.

[97] Geek for Geeks. *Autokey Cipher | Symmetric Ciphers*. https://www.geeksforgeeks.org/autokey-cipher-symmetric-ciphers/. 2023.

[98] Wikipedia contributors. *Tabula recta — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Tabula_recta&oldid=1098367810. 2022.

[99] Wikipedia contributors. *CRISPR — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=CRISPR&oldid=1148671998. 2023.

[100] Wikipedia contributors. *Receptor (biochemistry) — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Receptor_(biochemistry)&oldid=1128088142. 2022.

[101] Wikipedia contributors. *Spike protein — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Spike_protein&oldid=1136358529. 2023.

[102] Wikipedia contributors. *Monocryl — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Monocryl&oldid=1089283964. 2022.

[103] Wikipedia contributors. *Leidenfrost effect — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Leidenfrost_effect&oldid=1150669104. 2023.

[104] Wikipedia contributors. *Potassium nitrate — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Potassium_nitrate&oldid=1150285002. 2023.

[105] Wikipedia contributors. *Genomics — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Genomics&oldid=1137582695. 2023.

[106] Wikipedia contributors. *Atmospheric pressure — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Atmospheric_pressure&oldid=1149513099. 2023.

[107] Wikipedia contributors. *Troposphere — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Troposphere&oldid=1148830274. 2023.

[108] Wikipedia contributors. *Locust — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Locust&oldid=1150428936. 2023.

[109] J. Draper. "The Direction of Desert Locust Migration". *Journal of Animal Ecology* 49 (1980). https://doi.org/10.2307/4238, pp. 959–974.

[110] World Meteorological Organization and Food and Agriculture Organization of the United Nations. *Weather and Desert Locusts*. https://library.wmo.int/doc_num.php?explnum_id=3213. WMO and FAO, 2016.

[111] Wikipedia contributors. *St. Elmo's fire — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=St._Elmo%27s_fire&oldid=1151732630. 2023.

[112] Wikipedia contributors. *Corona discharge — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Corona_discharge&oldid=1136058839. 2023.

[113] Wikipedia contributors. *Ionization — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Ionization&oldid=1140127035. 2023.

[114] Wikipedia contributors. *Electromagnetism — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Electromagnetism&oldid=1149513663. 2023.

[115] Wikipedia contributors. *Plasma (physics) — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Plasma_(physics)&oldid=1149004458. 2023.

[116] Wikipedia contributors. *Glow discharge — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Glow_discharge&oldid=1134203238. 2023.

[117] Wikipedia contributors. *Stratosphere — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Stratosphere&oldid=1148830346. 2023.

[118] Wikipedia contributors. *Tesla coil — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Tesla_coil&oldid=1151818860. 2023.

[119] Wikipedia contributors. *Fischer projection — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Fischer_projection&oldid=1342035193. 2026.

[120] Wikipedia contributors. *Xenon hexafluoride — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Xenon_hexafluoride&oldid=1312857948. 2025.

[121] Wikipedia contributors. *Noble gas — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Noble_gas&oldid=1148164891. 2023.

[122] Wikipedia contributors. *Valence electron — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Valence_electron&oldid=1149655678. 2023.

[123] Wikipedia contributors. *EC — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=EC&oldid=1151770277. 2023.

[124] Wikipedia contributors. *EC50 — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=EC50&oldid=1151175482. 2023.

[125] Wikipedia contributors. *Electrical resistivity and conductivity — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Electrical_resistivity_and_conductivity&oldid=1148256130. 2023.

[126] Wikipedia contributors. *PH — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=PH&oldid=1151784533. 2023.

[127] Wikipedia contributors. *Phosphate test — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Phosphate_test&oldid=1105669068. 2022.

[128] National Drug Intelligence Center. *Methamphetamine Drug Threat Assessment*. https://www.justice.gov/archive/ndic/pubs13/13853/product.htm. 2005.

[129] Wikipedia contributors. *Petroleum — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Petroleum&oldid=1151964790. 2023.

[130] Wikipedia contributors. *Hydrocarbon — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Hydrocarbon&oldid=1148876895. 2023.

[131] Wikipedia contributors. *Ectoplasm (cell biology) — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Ectoplasm_(cell_biology)&oldid=981146216. 2020.

[132] Wikipedia contributors. *Cell nucleus — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Cell_nucleus&oldid=1150910432. 2023.

[133] Wikipedia contributors. *Epidermis (botany) — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Epidermis_(botany)&oldid=1102485269. 2022.

[134] Wikipedia contributors. *Spectrometer — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Spectrometer&oldid=1147343603. 2023.

[135] Wikipedia contributors. *Chromatophore — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Chromatophore&oldid=1071301210. 2022.

[136] Wikipedia contributors. *Holism — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Holism&oldid=1183891676. 2023.

[137] A. M. Owen. "Thousands of coma patients may be conscious but we're ignoring them, says pioneering neuroscientist". *ScienceFocus* (2022). https://www.sciencefocus.com/the-human-body/comas-conscious-communicate.

[138] J. Claassen and B. L. Edlow. "Some People Who Appear to Be in a Coma May Actually Be Conscious". *Scientific American* (2022). https://www.scientificamerican.com/article/some-people-who-appear-to-be-in-a-coma-may-actually-be-conscious/.

[139] Wikipedia contributors. *Cofactor (biochemistry) — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Cofactor_(biochemistry)&oldid=1182640897. 2023.

[140] B. Alberts, A. Johnson, and J. L. et al. *Molecular Biology of the Cell*. https://www.ncbi.nlm.nih.gov/books/NBK26873/. Garland Science, 2002.

[141] E. Svoboda. "Cellular Life, Death and Everything in Between". *Quanta Magazine* (2019). https://www.quantamagazine.org/cellular-life-death-and-everything-in-between-20190708.

[142] K. Meyl. "DNA and cell resonance: magnetic waves enable cell communication". *DNA and cell biology* (2012). https://doi.org/10.1089/dna.2011.1415.

[143] L. E. Team. *Hair DNA – Testing, Forensic Analysis and Tools*. https://laboratoryinfo.com/hair-dna-testing-forensic-analysis/. 2022.

[144] Wikipedia contributors. *Electron microscope — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Electron_microscope&oldid=1187540626. 2023.

[145] Wikipedia contributors. *Enzyme inhibitor — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Enzyme_inhibitor&oldid=1186613074. 2023.

[146] Z. Julier et al. "Promoting tissue regeneration by modulating the immune system". *Acta Biomaterialia* (2017). https://www.sciencedirect.com/science/article/pii/S1742706117300661.

[147] BBC. *Solids, Liquids and Gases*. https://www.bbc.co.uk/bitesize/topics/z242m39/articles/z726m39. 2023.

[148] P. Parkhe and S. Verma. "Evolution, Interspecies Transmission, and Zoonotic Significance of Animal Coronaviruses". *Front Vet Sci.* (2021). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8560429/.

[149] Wikipedia contributors. *Alpha helix — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Alpha_helix&oldid=1186861581. 2023.

[150] Wikipedia contributors. *Folding (chemistry) — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Folding_(chemistry)&oldid=1165781456. 2023.

[151] Wikipedia contributors. *Covalent bond — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Covalent_bond&oldid=1187593497. 2023.

[152] J. Clark. *The Structure of Proteins*. https://chem.libretexts.org/Bookshelves/Biological_Chemistry/Supplemental_Modules_(Biological_Chemistry)/Proteins/Protein_Structure/The_Structure_of_Proteins. 2023.

[153] Wikipedia contributors. *Naloxone — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Naloxone&oldid=1237245895. 2024.

[154] Wikipedia contributors. *Planck relation — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Planck_relation&oldid=1314768744. 2025.

[155] Wikipedia contributors. *Photon — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Photon&oldid=1364822874. 2026.

[156] Wikipedia contributors. *Photon energy — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Photon_energy&oldid=1358244359. 2026.

[157] Wikipedia contributors. *Planck constant — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Planck_constant&oldid=1360074938. 2026.

[158] Wikipedia contributors. *Kilogram — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Kilogram&oldid=1359305432. 2026.

[159] Wikipedia contributors. *Remote viewing — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Remote_viewing&oldid=1364702909. 2026.

[160] Wikipedia contributors. *Doppler effect — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Doppler_effect&oldid=1347967561. 2026.

[161] Wikipedia contributors. *Relativistic Doppler effect — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Relativistic_Doppler_effect&oldid=1362555364. 2026.

[162] Wikipedia contributors. *Hawking radiation — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Hawking_radiation&oldid=1364563195. 2026.

[163] Wikipedia contributors. *Relativistic dynamics — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Relativistic_dynamics&oldid=1315832323. 2025.

[164] Wikipedia contributors. *Special relativity — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Special_relativity&oldid=1365390777. 2026.

[165] Wikipedia contributors. *Lagrange multiplier — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Lagrange_multiplier&oldid=1363206730. 2026.

[166] Wikipedia contributors. *Inner product space — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Inner_product_space&oldid=1340828148. 2026.

[167] Wikipedia contributors. *Gradient — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Gradient&oldid=1355918370. 2026.

[168] D. Pham and Z. Musielak. "A Review of Lagrangian Formalism in Biology: Recent Advances and Perspectives". *arXiv* (2024). DOI: 10.48550/arXiv.2408.10834.

[169] P. García-Risueño, P. Echenique, and J. L. Alonso. "Exact and efficient calculation of lagrange multipliers in biological polymers with constrained bond lengths and bond angles: Proteins and nucleic acids as example cases". *Computational Chemistry* (2011). DOI: 10.1002/jcc.21885.

[170] Wikipedia contributors. *Nucleation — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Nucleation&oldid=1354603730. 2026.

[171] Wikipedia contributors. *Protein crystallization — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Protein_crystallization&oldid=1361282525. 2026.

[172] W.-F. Xue. "Nucleation: The Birth of a New Protein Phase". *Biophysical Journal* (2015). DOI: https://doi.org/10.1016/j.bpj.2015.10.011.

[173] A. Esmon. *The Physics of Ice: It All Begins with Nucleation*. www.thermofisher.com/blog/biobanking/the-physics-of-ice-it-all-begins-with-nucleation/. 2014.

[174] F. Franks et al. "Ice nucleation and freezing in undercooled cells". *Cryobiology* (1983). DOI: https://doi.org/10.1016/0011-2240(83)90018-4.

[175] EarthDate. *Frogsicles*. https://www.earthdate.org/episodes/frogsicles. 2025.

[176] C. F. A. Australia. *Can Blood Freeze?* https://cprfirstaid.com.au/can-blood-freeze/. 2025.

[177] Britannica. *Cryonics*. https://www.britannica.com/science/cryonics.

[178] B. S. Focus. *The hidden flaw in cryonics that could ruin your shot at living forever*. https://www.sciencefocus.com/the-human-body/freezing-brain-back-to-life. 2025.

[179] Wikipedia contributors. *Rapid amplification of cDNA ends — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Rapid_amplification_of_cDNA_ends&oldid=1301970372. 2025.

[180] Nature. *Whole genome amplification articles from across Nature Portfolio*. https://www.nature.com/subjects/whole-genome-amplification. 2026.

[181] Wikipedia contributors. *Methadone — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Methadone&oldid=1366705786. 2026.

[182] Wikipedia contributors. *Faraday cage — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Faraday_cage&oldid=1362219557. 2026.

[183] S. American. *Is Our Future Really Written in Our Genes?* https://www.scientificamerican.com/blog/observations/is-our-future-really-written-in-our-genes/. 2019.

[184] BBC. *How genetics determine our life choices*. https://www.bbc.com/future/article/20230509-how-genetics-determine-our-life-choices. 2023.

[185] Wikipedia contributors. *Epigenetics — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Epigenetics&oldid=1361169610. 2026.

[186] M. P. Institute. *Do your genes define who you are?* https://www.mpi.nl/blog/do-your-genes-define-who-you-are. 2025.

[187] Wikipedia contributors. *Iron(III) oxide — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Iron(III)_oxide&oldid=1356352485. 2026.

[188] A. Palleja, K. H. Mikkelsen, and S. K. Forslund. "Recovery of gut microbiota of healthy adults following antibiotic exposure". *Nat. Microbiol.* (2018). DOI: 10.1038/s41564-018-0257-9.

[189] N. L. of Medicine. *1-Chloroheptane*. https://pubchem.ncbi.nlm.nih.gov/compound/1-Chloroheptane.

[190] B. T. S. Team. "Application of 1-Chloroheptane in Analytical Chemistry: Detailed Notes and Protocols". *BenchChem* (2026).

[191] Wikipedia contributors. *Chromatography — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Chromatography&oldid=1360629607. 2026.

[192] Wikipedia contributors. *Spectroscopy — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Spectroscopy&oldid=1365857712. 2026.

[193] Wikipedia contributors. *Halogenation — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Halogenation&oldid=1334448434. 2026.

[194] Wikipedia contributors. *Halogen — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Halogen&oldid=1359713066. 2026.

[195] Wikipedia contributors. *Alkane — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Alkane&oldid=1361137291. 2026.

[196] Wikipedia contributors. *Haloalkane — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Haloalkane&oldid=1359424508. 2026.

[197] Wikipedia contributors. *Cyclophosphamide — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Cyclophosphamide&oldid=1341984314. 2026.

[198] Wikipedia contributors. *Myocarditis — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Myocarditis&oldid=1366113116. 2026.

VIGENÈRE AND CAESAR CIPHERS

```python
import math

def caesar_encrypt(plaintext,shift):
  ciphertext = ''
  for i in range(len(plaintext)):
    letter = plaintext[i]

    if (letter.isupper()): #upper case letters
      ciphertext += chr((ord(letter) + shift - ord('A')) % 26 + ord('A'))
    if (letter.islower()): #lower case letters
      ciphertext += chr((ord(letter) + shift - ord('a')) % 26 + ord('a'))
  return ciphertext


plaintext = input('Plaintext (without spaces): ')
shift = int(input('Shift value: '))

print('Ciphertext:', caesar_encrypt(plaintext,shift))

#Output
>> Plaintext (without spaces): HELLOWORLD
>> Shift value: 7
>> Ciphertext: OLSSVDVYSK
```

Listing 1. Encryption of the Caesar Cipher

```
#create tabula recta
tabularecta = []
for r in range(0,26):
  position = 0
  row = []
  for column in range(0,26):
    row.append(chr(r + ord('A') + position))
    position += 1
    if position > (25 - r):
        position -= 26
    tabularecta.append(row)

def repeat_to_length(string, wanted):
    return (string * (wanted//len(string) + 1))[:wanted] #repeats string until wanted length

def vigenere_cipher(plaintext, keyword):
  keyword_repeated =  repeat_to_length(keyword, len(plaintext)) #repeat keyword until matches length of plaintext
  ciphertext = []

  for i in range(len(plaintext)):
    keyword_index = ord(keyword_repeated[i]) - ord('A')
    letter = plaintext[i]
    letter_index_lower = ord(letter) - ord('a')
    letter_index_upper = ord(letter) - ord('A')

    if (letter.isupper()): #upper case letters
      enciphered_letter = tabularecta[keyword_index][letter_index_upper]
      enciphered_letter = chr(((letter_index_upper + keyword_index) % 26) + ord('A'))
      ciphertext.append(enciphered_letter)

    if (letter.islower()): #lower case letters
      enciphered_letter = tabularecta[keyword_index][letter_index_lower]
      enciphered_letter = chr(((letter_index_lower + keyword_index) % 26) + ord('a'))
      ciphertext.append(enciphered_letter)
  return ''.join(ciphertext)

plaintext = input('Enter plaintext (without spaces):' )
keyword = input('Enter keyword (upper case):' )

print('Ciphertext:' ,vigenere_cipher(plaintext,keyword))

#Output
```

```
>> Enter plaintext (without spaces): HelloWorld
>> Enter keyword (upper case): CAESAR
>> Ciphertext: JepdoNqrpv
```

Listing 2. Encryption of the Vigenère Cipher

LIST OF DIALOGUES

Below you will find all the dialogue that has been analysed in this meta, per episode. If you click on the episode title (in purple), then you will directly be directed to that section in the book.

1X01 PILOT

Liz: *"Every small town has a story, but my hometown has a legend. Roswell was a sleepy cowboy settlement in postwar America, full of farmers and military men, until, one day, something extraordinary happened. Or so the legend goes. Ever since, UFO enthusiasts have flooded in, searching for some cosmic phenomenon to prove we're not all alone in the universe."*

Alex: *"I'm serious. My chemical engineers found high levels of phenyl-2-propanone around your Airstream."*

1X02 SO MUCH FOR THE AFTERGLOW

Liz: *"I'm a scientist, Max. It's what I do. I got a grant for my vascular regeneration study."*

1X03 TEARIN' UP MY HEART

Liz: *"Chronic low-level anxiety keeps the hypothalamic pituitary adrenal axis activated"*

Liz: *"Prediction: Max's ability to manipulate electromagnetic energy gives him the power to heal and also to harm."*

Liz: *"I've established your baseline. Now I'm monitoring your electromagnetic output and how it correlates with... emotional triggers."*

Liz: *"Okay, are you all capable of electrostatic discharge... Releasing energy intentionally?"*

Liz: *"In nature, there are animals... Plants, even... That do variations of this."*

Liz: *"So if you can heal with electricity, can you also harm?"*

1X04 WHERE HAVE ALL THE COWBOYS GONE

Kyle: *"They're running a study on angiogenesis."*

Kyle: *"We lost transcutaneous pacing."*

1x05 Don't speak

Liz: *"But the work that you're doing with vessel regrowth is amazing. I mean, a few years ago, people never could have imagined this kind of progress in stem cell research. I mean, it's borderline science fiction. Look, this kind of science, healing what others believe to be ir- reparable, is important to me. It's all I've ever wanted to do."*

1x07 I saw the sign

Everett: *"Commencing the midline laparotomy."*

1x08 Barely breathing

Liz: *"I was working on this serum, a way to strip Isobel of her powers. I figured it might be a way to help people. It eliminates the electromagnetic charge around Max's squamous epithelial cells. So I hypothesize that the charge is connected to their powers. So, for all intents and purposes, it could render the aliens powerless. Human. But, I mean, it could do a lot worse, too. I don't know how much of their function is connected to this charge, so... I don't exactly have an alien mouse colony to test."*

Liz: *"Okay, so we'll occlude the coronary artery of half the mice, reperfuse it..."*

Kyle: *"The serum is attacking her cytoplasm. Her cells are rotting."*

Liz: *"I combined synthetic exothermic enzymes and melted down my jewelry to create a solution that should manipulate the properties of the membrane."*

1x10 I don't want to miss a thing

Alex: *"Oh, you know. Just a fragment of a 70-year-old UFO. Your dad left it for me in the cabin. Some of the symbols in these letters match the symbols on this glass. I wonder if it correlates to a specific mathematical cipher. I'll start with the Fibonacci sequence, and then maybe Shor's algorithm."*

2x01 Stay (I missed you)

Liz: *"It could be some sort of organic phosphorus?"*

2x02 Ladies and gentlemen we are floating in space

Liz: *"Human tissue can obviously regenerate from stem cells. With the right methodology, I could use your blood to make adjustments for alien physiology. I have to monitor exactly when cell degradation begins, down to the second."*

2x03 Good mother

Liz: *"Three years ago, I hypothesized that if I introduced a rare protein to destroyed stem cells, they'd regenerate. And I was right. My team in Denver brought dead cells back to life. Rat cells, but, I mean, still, the applications are immeasurable... until our study got shut down. They said it was for ethics reasons, but I think it's because it threatened big pharma. Then a few weeks ago, Kyle found that the pods contain a sort of alien cousin to my regenerative protein. When Max healed Rosa, his electric charge amplified the process. If we can replicate that, then, we can accelerate his recovery, so, I am testing out pig hearts to see..."*

2x04 What if God was one of us

Liz: *"Once Max is healthy, we could use this genome machine to target cellular apoptosis. I mean, we could craft polymerase sequencing in human DNA. We don't have to stop. We have no boards, no restrictions."*

2x06 Sex and candy

Kyle: *"You're here for the surgical separation of craniopagus twins?"*

2x09 The dinner

Liz: *"Hypothesis. The specimens display resistance to earthly infectious diseases due to alien hereditary immunity. Hypothesis. Pooled immunoglobulin harvested from an alien donor can be transferred to a human recipient. Human subjects exposed to specific sequencing of alien DNA will develop immunity to corresponding diseases."*

2x10 American woman

Liz: *"Do you know what butyricol is? Worth a shot. It's this chemical I found in my friend's tox screen. I have never heard of it."*

2x11 Linger

Diego: *"So, what are you doing with this worm behind my back?"*

Liz: *"Oh, administering a lethal dose of radiation to observe a particular neoblast. Don't be jealous."*

Liz: *"It's a regenerative master cell. The only one with this protein. I mean, this particular cell was able to multiply, diversify and reanimate my worm."*

Diego: *"Wait. So observing the master gets you the underlying mechanisms of tissue regeneration."*

Liz: *"Then I apply the mechanism to human tissue, and irreversible injury and degradation become distant memory."*

Diego: *"I bet butyricol uses inducible diphtheria toxins to stun the memory expression neurons into paralysis. See, if we can counteract that, we can get your friend her memories back."*

Liz: "Would it boost GABA in DLPFC?"

Diego: "See, no, you're thinking working memory, not recovery. Think traumatic memory."

Diego: "You're making calcium carbonate. Is this about regenerating worms?"

2x13 Mr Jones

Michael: "Ooh, the synthetic nucleotide excision repair genomogenate? Mm-hmm. We're lucky. You're only part alien, otherwise there wouldn't have been enough left of you for her to save."

3x01 Hands

Heath: "Oh, please, Dr. Roswell, don't cut me. Cytosine promises he will never hook up with adenine ever again."

Heath: "Hey, you know what? Remove a plasmid from a bacteria, I'm your guy. I'm not Ethan Hunt. I still have student loans."

3x03 Black hole sun

Liz: "My coworker at Genoryx reminded me that bacterial endospores can withstand even the most extreme environmental conditions. And he did it with rum."

3x05 Killing me softly with his song

Paramedic: "Systolic BP's 170 and rising. Pushing 100 milligrams of lidocaine."

Heath: "You are simultaneously presenting symptoms of vascular degeneration and African trypanosomiasis."

3x06 Bitter sweet symphony

Eduardo: "Laparoscopic technology is invasive."

3x07 Goodnight Elizabeth

Liz: "Hypothesis: alien cells use hydrolyse component polymers to attach themselves to host cells."

Michael: "You don't just hear things. Your ears are detecting subatomic vibrational frequencies."

Michael: "All matter in the universe emits signature sound wavelengths: rocks, uh, wind, fire, plants, even people."

Rosa: "Wait, is that how I was able to find Maria in her memories?"

Michael: "Neuronal activity produces brain waves oscillating at a high beta frequency. Your noggin picked up on those. That's how you found her. But I wonder, can you detect resting frequency gradients?"

3x08 Free your mind

Liz: *"The nanoblades will now be combined with the newly modified alien serum and infiltrate the Jones-Maria cell cluster for a cellular separation test."*

Liz: *"I found this research explaining how sound frequency can break plaque bonds in Alzheimer's patients."*

3x09 Tones of home

Kyle: *"I suggest that you have your team relook at the endoplasmic reticulum for traces of the paralytic."*

Alex: *"Working theory is a take on the Vigenère cipher with a twist."*

3x10 Angles of the silence

Liz: *"Heath's still working under the assumption that CRISPR will splice through binding proteins."*

Heath: *"Oh, it's okay. Liz won't mind if I borrow some of her CRISPR chemicals."*

Liz: *"Maybe I could use that to reason with him. Heath still thinks CRISPR is the answer. There is only one place that he would go."*

Liz: *"Bandelier University keeps CRISPR in their biolab."*

Liz: *"CRISPR isn't gonna get you all the way there."*

Heath: *"Jones thinks CRISPR has a part to play in his endgame. I can buy you some time."*

3x12 I ain't goin' out like that

Heath: *"Huh. Binding receptors are multiplying. They're effectively stopping the peplomers in Jones' cells from reattaching to Max's DNA. Oh. But at quite the cost."*

Isobel: *"4-0 Monocryl sutures ready to roll."*

Heath: *"Uh, U11, U12, U5. We have all of the pieces. We're just missing the mystery protein: UX. Man, it's got to be the key to repairing the DNA while making it inaccessible to Jones' tether, so, close, but no cigar."*

4x01 Steal my sunshine

Liz: *"Based on week seven's quiz results, we're going to do a refresh on the Leidenfrost Effect."*

Liz: *"All right. This is potassium nitrate. This is common table sugar. Apart... nada. But together..."*

Kyle: *"Got some new genomics software you might like to test drive. Or maybe we can see what else us interstellar internists can discover by examining Jones?"*

Alex: *"Okay, so, now everything here is yours, except for the level three clearance laptops, um, and the sat link phone and the hard drives."*

Alex: "I know that studying weather balloons doesn't sound like it's saving the world, but its research harbors some pretty cutting-edge stuff."

Eduardo: "And have you crunched the analytics for your weather balloons?"

Alex: "Yeah. Heat, uh, is up 1.3 degrees an hour for the last three over Roswell at large. Even though barometric pressure and wind direction dictates it should be falling. Not to mention that the ground conditions don't match up, that humidity is up 23% in the troposphere."

Kyle: "Is that why you had me studying locusts that had shifted their migrational course?"

Alex: "You think they were affected by the weather?"

Eduardo" "Potentially."

Dallas: "Yeah, I've been honing my connection to water lately. And I'm feeling this weird... shift in atmospheric pressure tonight."

4x02 Fly

Liz: "Perfect timing. I mean, test after test confirmed it's just a case of St. Elmo's fire. Scientific St. Elmo's fire... It's an electromagnetic corona discharge that glows due to high levels of plasmic voltage that rip apart molecules in the stratosphere."

Liz: "It's Earth science. It's not alien."

Liz: "We saw green streaks and shimmering rain because that is how our eyes perceive ionized protons."

Liz: "Except locusts aren't native to New Mexico. They are grasshoppers that have been genetically mutated under the right environmental conditions. So maybe this locust was affected by the St. Elmo's fire."

Liz: "It's... Locusts, they operate on a hive mind. So if the St. Elmo's fire altered their genetic sequence, maybe it activated their swarm instinct."

Liz: "Yeah, but it's really just... nitrogen and oxygen reacting to atmospheric electricity. Ionization... Okay, no harm in taking a tiny detour to explain the secrets of the universe. It's kind of hard to reproduce in a lab, but it operates a little bit like a Tesla coil."

4x03 Subterranean homesick alien

Liz: "Eduardo, thank you. This machine is my favorite thing. It tests for everything: EC, pH, PO4-P. I'm about to ask it to do my taxes."

Liz: "All right, well, the alien plasma matches the balloons, but that's to be expected since the samples we took were from their car wheels. Usual levels of bacteria and fungi, EC is normal. Huh... High levels of petroleum hydrocarbon."

4x04 Dear mama

Isobel: "All right, uh, Dr. Dyson, why don't you tell me about this, uh, alien ectoplasm, hmm?"

Liz: *“Oh, I prefer the term ‘goo’, but it’s technically epidermal tissue.”*

Liz: *"Yeah. I rigged it so that it would locate the alien tech component in alien biology. I mean, at least I hope I did.”*

Liz: *“Kyle. What if the alien cells aren’t dead? What if they are only playing dead? Like an octopus does when it-it’s scared and it turns into coral.”*

Kyle: *“You’re talking chromatophores.”*

4x06 Kiss from a rose

Liz: *“The regenerative protein that keeps you from getting sick is the same place where you derive your powers.”*

Liz: *“This virus was clearly built to replicate inside of a host, but it’s not replicating in this sample.”*

Shivani: *“Which means your friend can’t spread it.”*

Liz: *"Which is why I tipped that machete in Bonnie’s virus. Then I added a chemical accelerant to speed things up a little bit."*

Shivani: *"The regenerative protein we’ve been looking for. It’s damaged, but it’s a start."*

4x07 Dig me out

Liz: *"Wait, so... we were looking for a way to attack damaged tissue. Right? [...] Okay, but fractals, they’re all over the place. In cell physiology, and in human anatomy. I mean, what if we used that same pattern?"*

Shivani: *"We might be better able to holistically mitigate damage."*

Liz: *"Not only that. We could regenerate the cells altogether. I mean, you have this new cryogenics company, right? An immersive chamber might be exactly what we need to undo the damage that has altered these cells. We could flood our subjects’ circulatory systems with bio-engineered organic compounds that are in a cryogenic..."*

Shivani: *"Pod"*

4x08 Missing my baby

Liz: *"At first, I thought this was a resurgence of your powers, like a vision of Alex manifesting itself. But your brain scans match up with your last checkup, post-cure, which means this is not a vision. Alex is alive. And he’s figuring out a way to communicate with you. We’ve seen versions of this before. You did it in the coma. Max talked to Rosa in the pod."*

Liz: *"The regenerative enzymes in the blood sample we brought back from New York isn’t healing the diseased cells in our subject."*

Shivani: *"What about using the bio synthetic compound that my team developed?"*

Liz: *"The fluid stimulates the proteins momentarily, but it isn’t potent enough to sustain their viability."*

Shivani: *"The viscous liquid we developed was formulated using the fluid from within [the pod]. It contains restorative properties identical to the ones that you’ve been researching."*

Liz: *"In gene-sequencing your work, I found micromolecular inconsistencies, and this crack explains why. Oxygen is changing the fluid's chemical composition. In order to continue testing, we need a pod that's in working condition."*

Rosa: *"I can hear the frequency of Alex's DNA in the hair, but... nowhere else."*

Rosa: *"Energy rises from the ground into your feet, and since consciousness is a tangible form of energy connecting us to all matter, astrophysicists believe that... meditation could open you up to speaking over universes, even dimensions."*

Shivani: *"Why atomize the fluid into a mist?"*

Liz: *"I made slight genetic modifications to accelerate enzyme inhibitors for swift cell regeneration. Molecules, they travel faster in a gaseous state than in liquid form."*

Shivani: *"If this cure eradicates the alien virus, could it be repurposed to fight off human illness?"*

Liz: *"Like a universal vaccine? Interspecies scaling at this level would be a huge leap."*

Shivani: *"What if we altered the protein receptors to perform some, not all, of their functions? If we folded the principle alpha helices and stabilized the covalent bonds between the amino acids..."*

4x09 Wild Wild West

Michael: *"We need to find the naloxone."*

Heath: *"Well, I can't be sure until I run some more tests, but... cold hands, fever, rash– it could be autoimmune. [...] Meaning that the mist could be causing her body to attack itself. I'm gonna give her some steroids. It should suppress the immune response. Well, at least, in normal circumstances."*

Heath: *"Well, the drug has latched on to Liz's nerve receptors. Somehow it's managed to trick her immune system into thinking that her own nerves are actually foreign bodies. Which would be an easy fix, if I had a way to camouflage alien proteins inside human biology. Unfortunately, NYU hasn't quite gotten there yet."*

4x10 Down in a hole

Rosa: *"Well, Heath said that what you're sensing might be similar to my abilities. Everything has a resonant frequency, right? And I'm guessing so does the liminal space that our friends are stuck in."*

Maria: *"Wait, so you're saying that my new powers might be picking up frequencies in other places, too?"*

Rosa: *"Exactly. And as you tap into the energy rippling off that you're able to get glimpses of our friends who went all Land of the Lost. It's something that the CIA tried to do, is bend time and space to spy on their enemies. They called it remote viewing. But they didn't have a real psychic."*

Liz: *"Do you understand the relativistic dynamics of Hawking's radiation? Can you convert constrained Lagrange multipliers?"*

Liz: *"This frog's nucleating proteins ensure the water in its blood freezes first. So the cells shrink, and it enters suspended animation."*

Shivani: *"An earthbound way of replicating what those alien pods do."*

Liz: *"But our Kermit here is freshly dead, so I have an electrical current that's pulsing through him. Same as you have to with Nicole. And it's gonna stimulate and simulate stasis. But with the right chemical combo, we'll breathe new life into those waiting cells."*

4X11 FOLLOW YOU DOWN

Liz: *"I can't remember the genome sequence to rapid amplification…"*

Kyle: *"I'm working on a hybrid methadone that'll help ease the withdrawal symptoms."*

Max: *"What are these equations, Liz?"*

Liz: *"I am charting the nucleotides in your [Max's] triple helix to see if I can fight fire with actual fire."*

Dallas: *"Wait, these old shipping containers are coated in lead. That means electromagnetic waves can't pass through them."*

Bonnie: *"What does that mean?"*

Dallas: *"When I first got inducted into the pod squad, Michael told me that our powers are electromagnetic."*

Shivani: *"Real life? Which is governed by physics, chemistry, biology. Equal and opposite reactions. Cause and effect. Our entire lives written into our very DNA before we even take a first step. One big chain reaction leading to one inevitable conclusion."*

Liz: *"I need ferric oxide, i.e. rust."*

Rosa: *"Liz figured out that resonate frequencies can break psychic tethers and, with four roommates, I've gotten kind of good at silencing them, so maybe."*

Rosa: *"The frequency's destabilizing."*

Alex: *"If this is an alternate dimension, the amount of energy that it would take to build it would border on atomic fission. And like nuclear power, it would have unintended consequences. Radiation."*

Michael: *"The gas mask."*

Alex: *"I felt nauseous as soon as I got here. Did you? Dallas, Bonnie? [...] Okay, so it's only humans. Which is why Theo didn't realize. [...] The gas mask helped with the nausea, but it-it couldn't stop this. [...] I think I'm dying."*

Liz: *"You were right not to trust me. The mist was synthesized from Bonnie's virus, and by reverse engineering the chemical structure, I was able to mimic her abilities and render you powerless."*

TWO SPARROWS IN A HURRICANE

Liz: *"'C7H15Cl.' What is that?"*

Kyle: *"Cyclophosphamide? The alkylating agent used in cancer treatments."*

Liz: *"Yeah. Methadone nap must've given me a brain cramp."*

Liz: *"Look, Kyle ran me through the elevated risk of stroke and myocarditis, and since nothing speaks to me like the science, we're safe. No more mist."*

Index

Symbols

A

B

C

D

E

F

G

H

I

L

M

N

P

R

S

T

V

W